

JSS 2 COMPUTER HARDWARE AND GSM REPAIRS LESSON NOTES

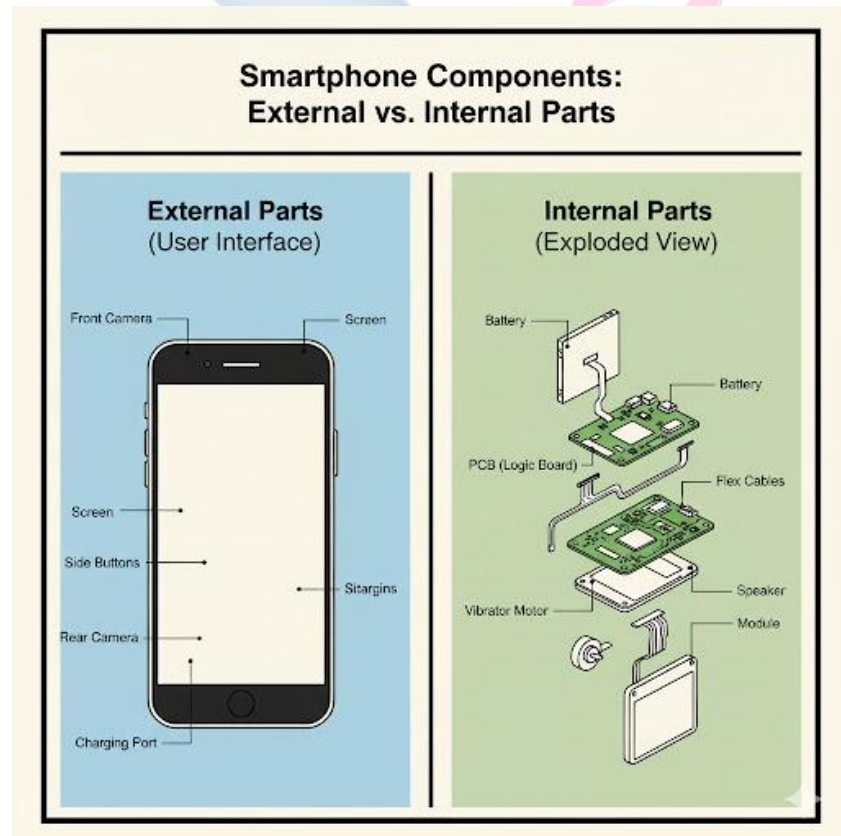
FIRST TERM

WEEK 1: REVIEW OF JSS 1 WORK

Content: Recap of phone parts, safety, and tools

Performance Objectives: By the end of this lesson, students should be able to:

1. List and identify at least five external parts and five internal components of a mobile phone.
2. Define "Electrostatic Discharge (ESD)" and state two ways to prevent it.
3. List five essential safety rules for a repair workshop.
4. Identify and state the function of ten (10) basic GSM repair tools.
5. Differentiate between a software fault and a hardware fault.



Lesson Content:

1. Recap of Mobile Phone Components

- **A. External Parts:** The parts the user can see and touch.
 - **Screen/Display:** Shows images and text (LCD or OLED).
 - **Casing/Housing:** The outer "body" that protects the phone.
 - **Buttons:** Power button, volume buttons, (formerly home button).
 - **Ports:** Charging port (e.g., USB-C), SIM card tray, (formerly headphone jack).
 - **Cameras:** Front and rear lenses.
 - **Openings:** Earpiece speaker, loudspeaker grill, microphone hole.
- **B. Internal Components:** The parts inside the phone.
 - **PCB (Printed Circuit Board):** The "motherboard" or "brain" of the phone. It's the green board that holds all the important chips.
 - **Battery:** Provides power to the phone (usually Lithium-ion).
 - **Flex Cables:** Flexible "ribbons" that connect different parts (e.g., connect the screen to the PCB).
 - **Vibrator Motor:** A small motor that spins an unbalanced weight to make the phone vibrate.
 - **Speakers & Microphone:** For sound.
 - **ICs (Integrated Circuits):** The small black "chips" soldered to the PCB that perform specific jobs (e.g., Power IC, CPU).

2. Review of Safety Procedures

Safety is the *most important* part of any repair.

- **A. Personal Safety:**
 - Wear safety goggles to protect your eyes from flying debris.
 - Work in a well-ventilated area to avoid breathing in fumes from solder.
 - Be careful with sharp tools and hot equipment (like a soldering iron).
- **B. Electrostatic Discharge (ESD) Safety:**

- **Definition:** ESD is the sudden flow of static electricity from your body into the sensitive components (like the PCB or ICs) of the phone. This *tiny* shock is too small for you to feel, but it can *destroy* the phone's internal parts.
- **Prevention:**
 1. **ESD Mat:** Work on a special rubber mat that grounds you.
 2. **ESD Wrist Strap:** Wear a wrist strap that connects you to the mat, safely discharging any static buildup from your body.
 3. **No Wool/Nylon:** Avoid wearing clothes that build up static.
 4. **Touch Metal:** Before touching any internal parts, touch a large metal object (like the table leg) to discharge yourself.

DON'T



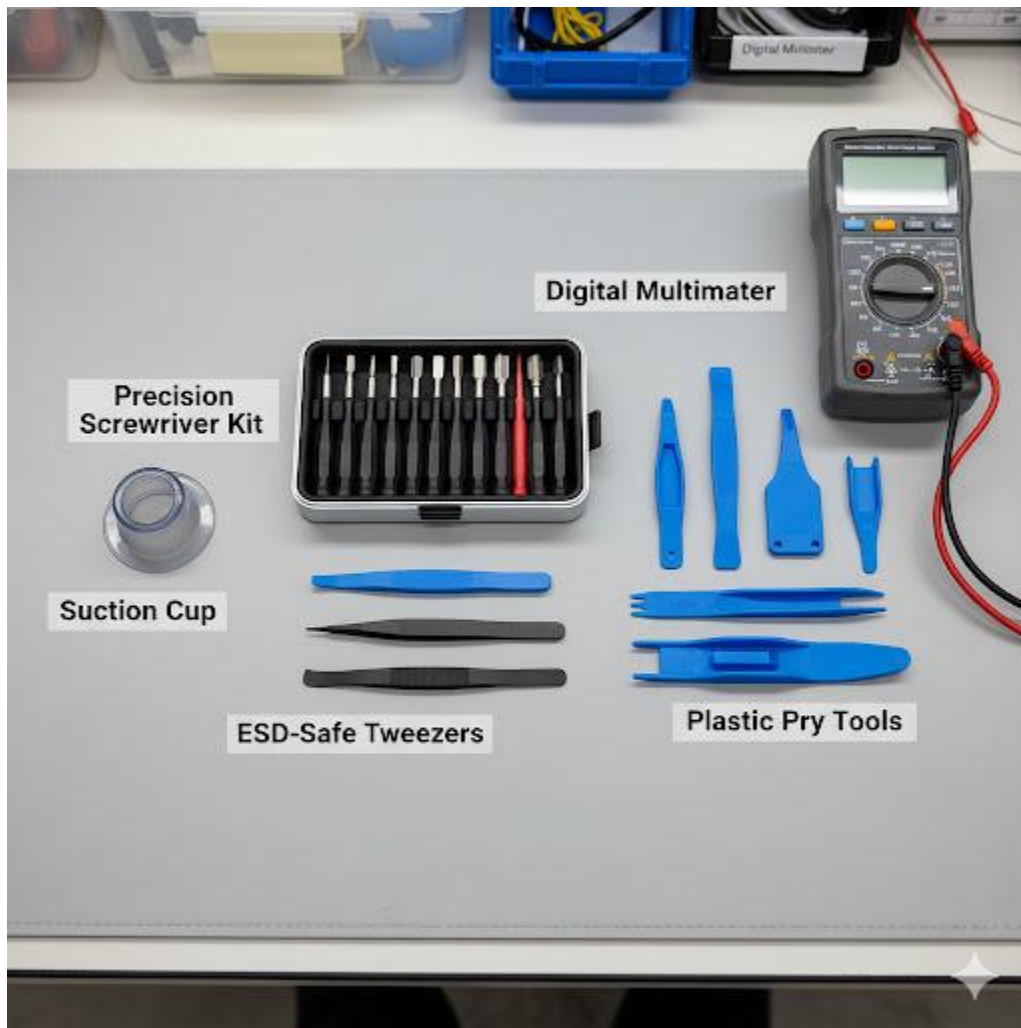
DO ✓



- **C. Battery Safety:**

- *Never* puncture, bend, or heat a lithium-ion battery. It can explode or catch fire.
- Always use plastic tools (not metal) when prying a battery out.

3. Review of Basic GSM Repair Tools



1. **Screwdriver Kit:** A set of small, precision screwdrivers (e.g., Torx, Pentalobe, Phillips) to open different phone models.
2. **Pry Tools (Plastic):** Spudgers, picks, and opening tools used to *gently* separate the casing without scratching it.
3. **Tweezers (ESD-Safe):** For picking up tiny screws and components.
4. **Suction Cup:** Used to lift the screen off the phone body.

5. **Multimeter:** The *most important* diagnostic tool. Used to measure voltage, current, and continuity (to check for breaks in a circuit).
6. **Soldering Iron:** Used to melt solder to join or remove components from the PCB.
7. **Hot Air Rework Station:** Uses hot air to melt solder, used for removing and replacing ICs.
8. **Magnifying Lamp:** A lamp with a magnifying glass to see tiny components clearly.

Evaluation:

1. What is the "brain" or "motherboard" of a phone called?
2. What is ESD, and why is it dangerous?
3. List two ways to prevent ESD.
4. Why should you *never* use a metal tool to remove a battery?
5. What is a "multimeter" used for?

Assignment:

1. Define "PCB" and "Lithium-ion Battery."
2. Draw and label five (5) external parts of a mobile phone.
3. Draw and identify five (5) tools from a basic GSM repair kit.
4. What is the difference between a software fault (e.g., phone is slow) and a hardware fault (e.g., broken screen)?
5. Write down three (3) important safety rules you must always remember.

WEEK 2: INTRODUCTION TO MOBILE CIRCUITS

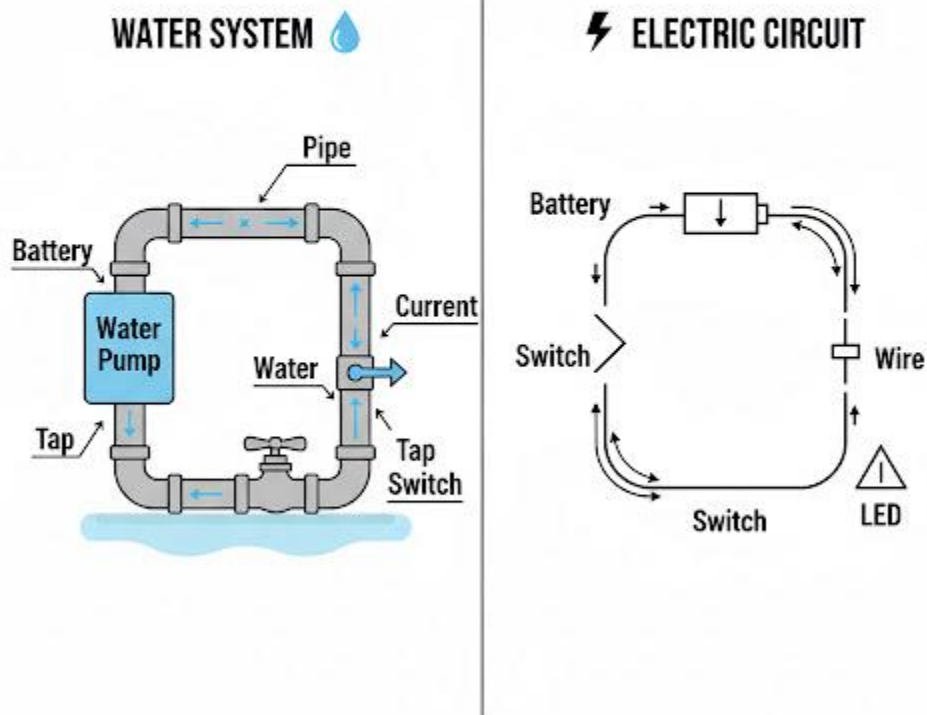
Topic: Introduction to Mobile Circuits

Content: Meaning of circuit; Simple phone circuit components and symbols.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "electrical circuit."
2. Differentiate between a "complete circuit" and an "open circuit."
3. Define "current," "voltage," and "resistance."
4. Identify the symbols for at least five basic electronic components.
5. Draw a simple circuit diagram.

CIRCUIT ANALOGY



Lesson Content:

1. What is an Electrical Circuit?

- **Definition:** A circuit is a **complete, closed path** that allows electricity (current) to flow. If the path is broken at any point, it is an "open circuit," and electricity cannot flow.
- **Analogy:** Think of it like a water pipe system.
 - The **Battery** is the **Water Pump** (provides pressure).
 - The **Wire** is the **Pipe** (carries the flow).
 - The **Current** is the **Water** (the "stuff" that flows).
 - A **Switch** is a **Tap** (turns the flow on or off).
 - A **Bulb** is a **Water Wheel** (a "load" that *uses* the flow to do work).
- **Mobile Circuit:** A phone's PCB is a very complex circuit board with thousands of tiny paths and components that direct electricity to the right places (e.g., to the screen, to the speaker).

2. Basic Electrical Concepts

1. **Current (I):** The *flow* of electricity (electrons) through the circuit. Measured in **Amperes (A)**.
2. **Voltage (V):** The *pressure* that pushes the current. Provided by the battery. Measured in **Volts (V)**. (e.g., a phone battery has ~3.7 Volts).
3. **Resistance (R):** The *opposition* to the flow of current. It "slows down" the electricity. Measured in **Ohms (Ω)**.

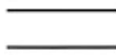
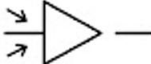
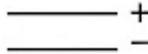
3. Basic Electronic Components and Symbols On a circuit diagram (a "map" of the circuit), we use symbols, not drawings.

Component	Function	Schematic Symbol
Battery	Provides DC Voltage (power)	
Wire	A path for current (conductor)	A straight line (---)
Switch	Opens or closes a circuit	
Resistor (R)	<i>Resists</i> or <i>limits</i> the flow of current	A zigzag line

Capacitor (C)	<i>Stores and releases</i> electricity (like a tiny, fast battery)	Two parallel lines (---
Diode (D)	Allows current to flow in <i>one direction</i> only	A triangle with a line (---
LED	(Light Emitting Diode) A diode that gives off light	(---
Transistor (Q)	A tiny switch or amplifier	
Ground (GND)	The common return path for the current	

4. Simple Circuit Diagram

COMMON CIRCUIT SYMBOLS

COMPONENT	SCHEMATIC SYMBOL
Resistor	
Capacitor	
LED (Light-Emitting Diode)	
Battery	

- A simple circuit consists of a **power source** (battery), a **path** (wire), and a **load** (what does the work, e.g., a bulb).
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- **How it works:** When the switch is closed, the path is complete. Current flows from the positive (+) terminal of the battery, through the switch, through the LED (which lights up), and back to the negative (-) terminal of the battery.

Evaluation:

1. What is an electrical circuit?
2. What is the difference between an "open circuit" and a "closed circuit"?
3. What is "voltage"? What is it measured in?
4. What component *resists* the flow of current?
5. What does a "diode" do?

Assignment:

1. Define Current, Voltage, and Resistance.
2. Draw the schematic symbols for:
 - (a) Battery
 - (b) Resistor
 - (c) Capacitor
 - (d) LED
3. What is the function of a "capacitor" in a circuit?
4. In the water pipe analogy, what is the battery, and what is the current?
5. (Practical) Draw a simple circuit diagram showing a battery, a switch, and a resistor, all connected in a loop.

WEEK 3: POWER SECTION OF A MOBILE PHONE

Topic: Power Section of a Mobile Phone

Content: Charging system, battery, and power IC functions

Performance Objectives: By the end of this lesson, students should be able to:

1. Trace the "power path" in a mobile phone.
2. Identify the components of the charging system.
3. Explain the function of the battery.
4. Explain the function of the Power IC (Integrated Circuit).
5. List two common faults related to the power section.

Lesson Content:

1. Overview of the Power Section The power section is the most important part of the phone. It is like the **heart and blood vessels** of the human body. It manages all the power, from charging the battery to distributing that power to every other component.

THE POWER PATH

PATH A: CHARGING



PATH B: POWERING ON



2. The Power Path (How Power Moves) The flow of power follows a specific path:

Path A: Charging the Phone (When plugged in)

1. **Charger (AC Adapter):** Takes 220V AC power from the wall.
2. **USB Cable:** Delivers 5V DC power to the phone.
3. **Charging Port:** The physical port on the phone.
4. **PCB (Circuit Board):** The path goes to the...
5. **Charging IC (Charge Controller):** A chip that *manages* the charging. It controls the voltage and current to safely charge the battery.
6. **Battery:** The 5V from the charger is used to charge the 3.7V-4.2V battery.

Path B: Powering the Phone (When in use)

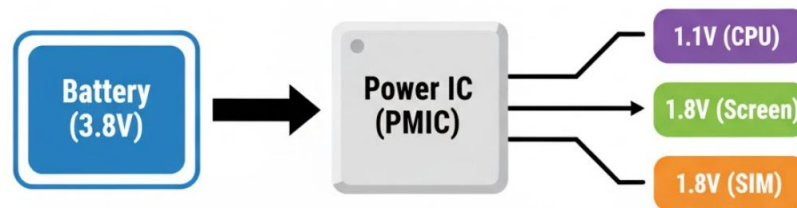
1. **Battery (3.7V - 4.2V):** Supplies power to the PCB.
2. **Power IC (PMIC - Power Management IC):** This is the "**main boss**" of power.
3. **CPU, Screen, etc.:** The Power IC distributes the *correct* voltage to all other parts.

3. Key Components of the Power Section

- **A. The Charging System:**
 - **Charging Port:** A common point of failure. It gets damaged by physical force (rough plugging/unplugging) or dirt/corrosion.
 - **Charging IC:** This chip is essential. If it fails, the phone will not charge (or will show "fake charging"). It also prevents the battery from overcharging.
- **B. The Battery (Lithium-ion):**
 - **Function:** To *store* electrical energy and release it as needed.
 - **Characteristics:** It is a DC power source, usually rated at 3.7V (nominal). When fully charged, it's ~4.2V. When "dead," it's ~3.5V or less.
 - **Battery Connector:** The point where the battery clips or plugs into the PCB.
- **C. The Power IC (PMIC):**
 - **Function:** This is a large, complex chip that acts as the phone's "power distribution box."
 - A phone's battery supplies one voltage (e.g., 3.8V), but different parts need different voltages:

- The CPU might need 1.1V.
- The Screen might need 1.8V and 3.0V.
- The SIM card might need 1.8V.
- The Power IC's job is to *receive* the 3.8V from the battery and *convert* it into all the different, smaller voltages that the other components need.

POWER IC (PMIC) FUNCTION



4. Common Power Section Faults

1. Phone Not Charging:

- *Causes:* Bad charger/cable (always check first!), dirty/damaged charging port, faulty Charging IC, or a dead battery.

2. Phone Not Turning On (Dead Phone):

- *Causes:* Deeply discharged/dead battery (most common), faulty Power button, faulty Power IC, or a "short circuit" on the PCB.

3. Battery Drains Very Fast:

- *Causes:* Old/weak battery, software bug (apps running in the background), or a "short circuit" on the PCB causing a component to "leak" power.

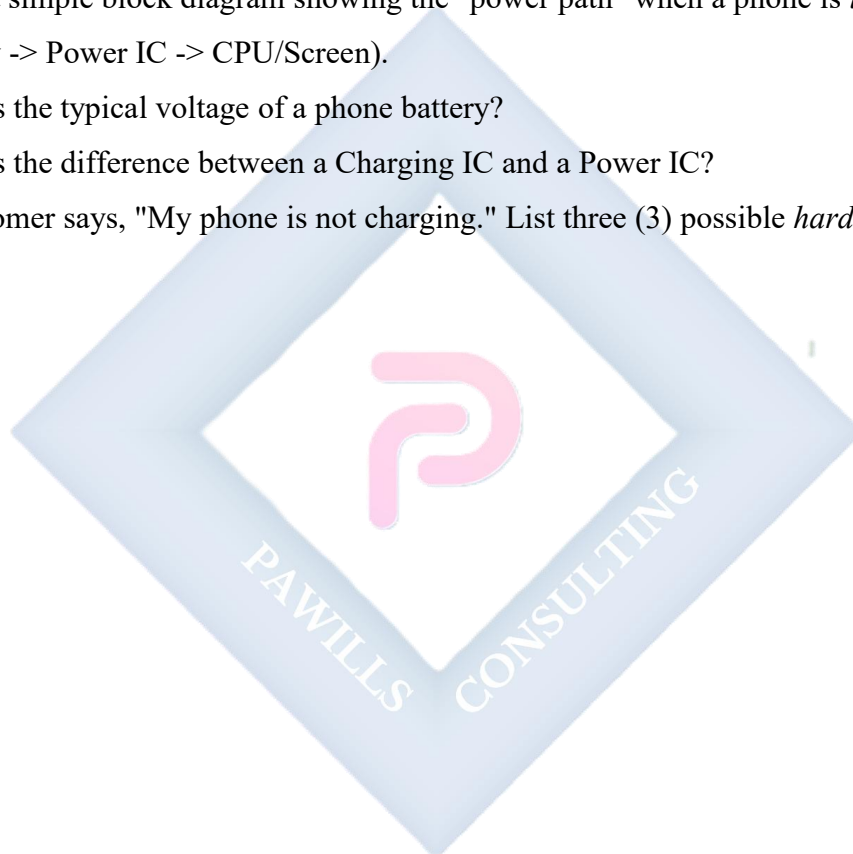
Evaluation:

1. What is the first component the electricity passes through after the charging port?
2. What is the "main boss" of power distribution in a phone?

3. What is the difference between AC and DC current?
4. What does the Power IC *do*? (e.g., "It takes one voltage and...").
5. List two common faults related to the power section.

Assignment:

1. Draw a simple block diagram showing the "power path" when a phone is *charging*. (e.g., Port -> Charging IC -> Battery).
2. Draw a simple block diagram showing the "power path" when a phone is *in use*. (e.g., Battery -> Power IC -> CPU/Screen).
3. What is the typical voltage of a phone battery?
4. What is the difference between a Charging IC and a Power IC?
5. A customer says, "My phone is not charging." List three (3) possible *hardware* causes.



Week 4: Display and Network Sections

Topic: Display and Network Sections

Content: Screen components; Antenna, network IC, SIM section; Troubleshooting display and network faults

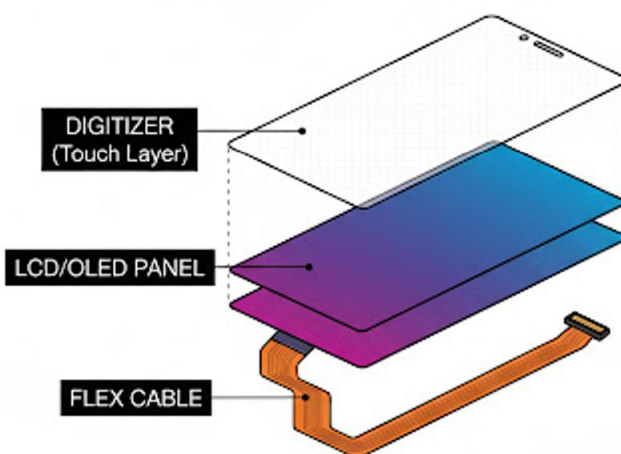
Performance Objectives: By the end of this lesson, students should be able to:

1. Identify the main components of the display section.
2. List two common faults related to the display and their possible causes.
3. Identify the main components of the network section.
4. Explain the function of the SIM card and the antenna.
5. List two common faults related to the network and their possible causes.

Lesson Content:

1. The Display Section This section is responsible for everything you see on the screen.

DISPLAY ASSEMBLY



- **A. Components:**

1. **LCD/OLED Screen:** The main display panel itself.

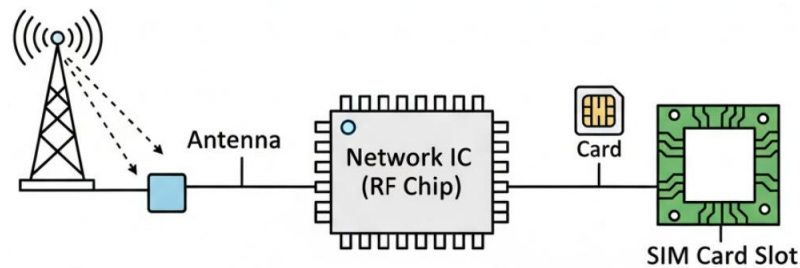
- **LCD (Liquid Crystal Display):** Needs a "backlight" to light it up.
- **OLED (Organic Light Emitting Diode):** Each pixel lights itself up (no backlight needed).
- 2. **Digitizer (Touchscreen):** A clear layer (often fused with the screen) that detects your touch.
- 3. **Flex Cable:** The flexible "ribbon" that connects the screen/digitizer to the PCB.
- 4. **Display Connector:** The socket on the PCB where the flex cable plugs in.
- 5. **Display IC / Backlight IC:** Chips on the PCB that control the image and the backlight.
- **B. Common Display Faults:**
 1. **Cracked Screen (No Touch):**
 - *Cause:* Physical impact (dropping the phone).
 - *Fault:* The Digitizer is broken.
 - *Repair:* The entire screen assembly must be replaced.
 2. **No Image / Black Screen (but phone makes sound):**
 - *Causes:*
 - (a) The flex cable is loose. (Repair: Re-seat the connector).
 - (b) The screen is internally damaged. (Repair: Replace the screen).
 - (c) The Backlight IC or Display IC on the PCB is faulty (Requires micro-soldering).
 3. **No Backlight (Image is very faint):**
 - *Cause:* The backlight circuit (coils, diode, IC) on the PCB has failed. This is a common fault after water damage.

2. The Network Section This section is responsible for making calls, sending texts, and using mobile data (2G, 3G, 4G, 5G).

- **A. Components:**
 1. **SIM Card:** (Subscriber Identity Module). A small chip that *identifies you* to the network (e.g., MTN, Glo).
 2. **SIM Card Slot:** The tray/reader that connects the SIM card to the PCB.

3. **Antenna:** A (usually) internal wire or plate that physically *sends and receives* the radio wave signals from the cell tower.
4. **Antenna Switch:** A component that switches between different signal bands (e.g., 3G and 4G).
5. **Network IC (RF Chip / PFO):** The "Radio Frequency" chip. This is the "brain" of the network section. It processes all the signals for sending and receiving.

NETWORK SECTION



- **B. Common Network Faults:**

1. **"No Network" / "Emergency Calls Only":**

- *Causes:*
 - (a) You are in a bad service area (always check!).
 - (b) The Antenna is disconnected or damaged.
 - (c) The Network IC is faulty (often from a hard drop).
 - (d) The phone's IMEI number is blacklisted (if stolen).

2. **"Insert SIM Card" / "No SIM":**

- *Causes:*
 - (a) The SIM card is inserted incorrectly.
 - (b) The SIM card is dirty or damaged. (Repair: Clean the gold contacts).
 - (c) The SIM Card Slot pins are bent or broken. (Repair: Replace the slot).

Evaluation:

1. What is the "digitizer"?
2. What is the difference between an LCD and an OLED screen?
3. What is the function of the SIM card?
4. What is the function of the Antenna?
5. A customer's phone turns on, but the screen is black. Name two possible causes.

Assignment:

1. Draw a simple block diagram of the display section (e.g., Screen/Digitizer -> Flex Cable -> Display IC -> PCB).
2. A customer's phone shows "No SIM Card." List three possible causes.
3. What is the "Network IC" (or RF Chip)?
4. A customer dropped their phone, and the glass is cracked, but the touch still works. What part is broken?
5. A customer dropped their phone, and the glass is fine, but the touch *does not* work. What part is broken? (Digitizer).

WEEK 6: SOUND AND CHARGING SECTIONS

Topic: Sound and Charging Sections

Content: Microphone, speaker, ringer, charging port; Common faults and repairs

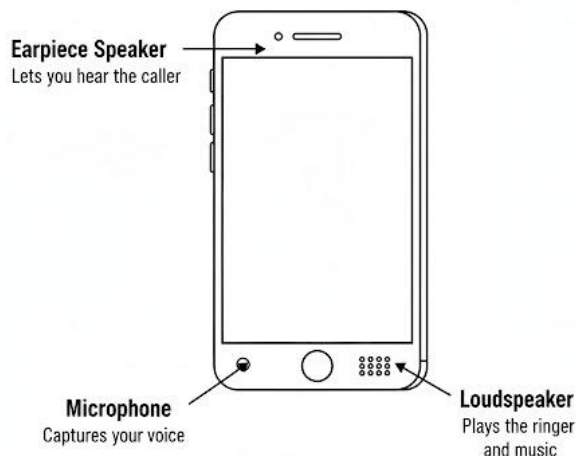
Performance Objectives: By the end of this lesson, students should be able to:

1. Differentiate between the earpiece speaker, the loudspeaker, and the microphone.
2. State the function of each of the three sound components.
3. List common faults and repairs for the sound section.
4. Identify the charging port as a common hardware fault.
5. Describe the steps to troubleshoot a "not charging" fault.

Lesson Content:

1. The Sound Section (Audio) A phone has three main sound components, controlled by an Audio IC.

AUDIO COMPONENTS



- **A. Earpiece Speaker (Receiver):**
 - **What it is:** The *small* speaker you hold to your ear during a call.
 - **Function:** To let you hear the *other person's voice* in a private call.
 - **Fault:** "I can't hear the person I'm calling."
 - **Repair:**
 1. Check the earpiece grill (hole) for dirt/debris. Clean it.
 2. Check the volume during the call.
 3. If faulty, the earpiece component is replaced.
- **B. Loudspeaker (Ringer):**
 - **What it is:** The *large* speaker, usually at the bottom or back of the phone.
 - **Function:** To play *loud* sounds: the ringer, music, video, and speakerphone.
 - **Fault:** "My phone doesn't ring," or "Music plays very quietly or sounds cracked."
 - **Repair:**
 1. Check the speaker grill for dirt. Clean it.
 2. Check software (volume settings, silent mode).
 3. If faulty, the loudspeaker module is replaced.
- **C. Microphone ("Mic"):**
 - **What it is:** A *tiny* component (often a small hole at the bottom of the phone).
 - **Function:** To *capture your voice* and send it to the other person.
 - **Fault:** "The other person can't hear me."
 - **Repair:**
 1. Check the microphone hole for dirt/debris. Clean it gently with a pin.
 2. Test with a voice recorder app.
 3. If faulty, the microphone (which is often soldered to the charging port flex or the main PCB) must be replaced.

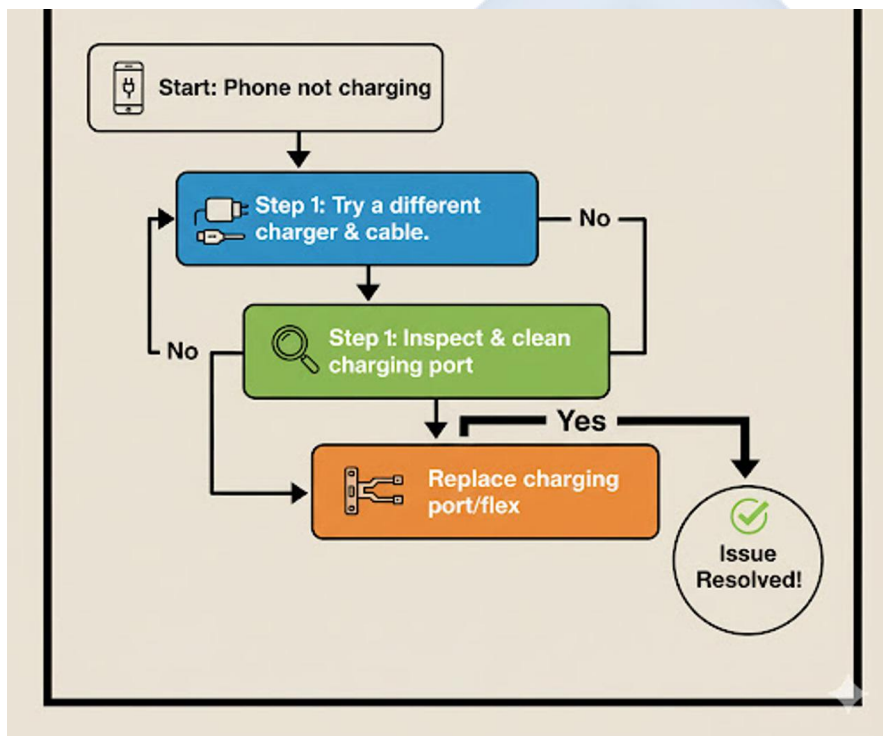
2. The Charging Section (Review) This section is responsible for getting power into the battery.

- **Components:** Charger, USB Cable, Charging Port, Charging IC, Battery.
- **The Charging Port:** This is the *most common* part to fail. It is a "mechanical" part that suffers from physical wear and tear every day.
- **Common Faults and Repairs:**

1. **Fault: "Phone is not charging"** (no charging symbol).

- **Troubleshooting Steps (in order):**

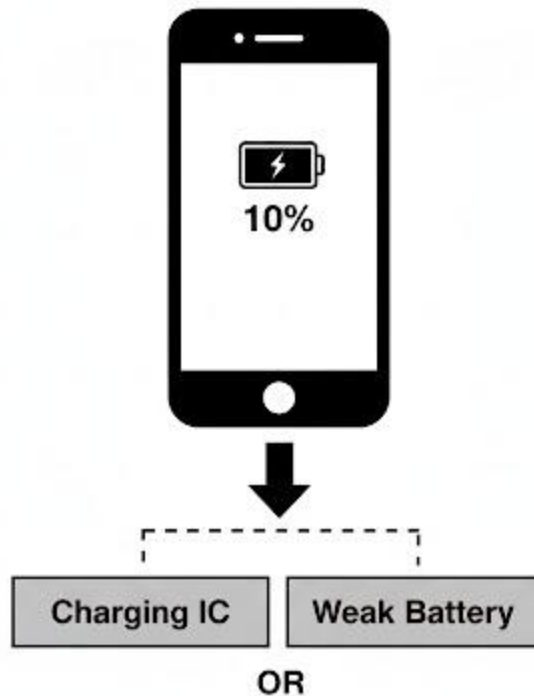
1. **Check the Obvious:** Try a *different charger* and a *different cable*.
2. **Visual Inspection:** Look inside the phone's charging port with a light. Is it full of lint or dirt?
3. **Clean the Port:** Use non-conductive tweezers or a wooden toothpick to *gently* scrape out any debris.
4. **Hardware Repair:** If it's still not charging, the port itself is likely damaged or has a broken connection. The charging port (or its flex cable) must be replaced.



2. **Fault: "Fake Charging"** (shows the charging symbol, but the percentage never goes up).

- **Causes:** This is *not* a port problem. It is usually a faulty **Charging IC** or a very **weak/old battery** that can no longer hold a charge.

FAKE CHARGING



Evaluation:

1. What is the function of the earpiece speaker?
2. What is the function of the microphone?
3. A customer says, "My phone doesn't ring, but I can hear people when I make a call." What part is faulty?
4. A customer says, "People can't hear me, but I can hear them." What part is faulty?
5. What is the *first* thing you should always do if a phone is not charging?

Assignment:

1. List the three main audio components of a phone and their functions.
2. What is the "Audio IC"?

3. Describe the step-by-step process for troubleshooting a "phone not charging" fault.
4. What is the likely cause of "fake charging"?
5. (Practical) Look at your own phone (or a family member's). Identify the earpiece speaker, the loudspeaker grill, the microphone hole, and the charging port.



WEEK 8: SOFTWARE UPDATES AND FLASHING

Topic: Software Updates and Flashing

Content: Meaning of flashing; Importance of updating OS; Tools and apps for flashing

Performance Objectives: By the end of this lesson, students should be able to:

1. Differentiate between "hardware" and "software."
2. Define "Operating System (OS)."
3. Define "flashing" (installing firmware).
4. State two reasons why it is important to update a phone's OS.
5. List the tools needed to flash a phone.

Lesson Content:

1. Hardware vs. Software (Review)

- **Hardware:** The *physical* parts of the phone you can touch (Screen, Battery, PCB, ICs).
- **Software:** The *instructions and programs* that run on the hardware (The OS, the Apps).

HARDWARE vs. SOFTWARE



2. What is an Operating System (OS)?

- **Definition:** The OS is the main, master software that manages all the phone's hardware and allows you to run other applications (apps).
- **Examples:**
 - **Android** (used by Samsung, Tecno, Infinix, etc.)
 - **iOS** (used *only* by Apple iPhones)
- **Firmware:** This is the specific version of the OS designed for your *exact* phone model.

BOOT LOOP FAULT



Stuck in a continuous restart cycle.

3. What are Software Updates?

- **Definition:** These are small patches or upgrades to your OS provided by the manufacturer (e.g., updating from Android 12 to Android 13).

- **Importance of Updating:**

1. **Security:** This is the *most important* reason. Updates "patch" security holes to protect your phone from viruses and hackers.
2. **New Features:** Updates often add new functions, new designs, and new apps.
3. **Bug Fixes:** They fix errors ("bugs") from the previous version, which can make your phone faster and stop it from crashing.

4. What is "Flashing"?

- **Definition:** "Flashing" is the process of manually installing or re-installing the entire *firmware* (the OS) on a phone.
- **Analogy:** It is like formatting a computer's hard drive and re-installing Windows. It erases *everything* and installs a fresh copy.
- **When is Flashing Used? (Software Faults)**
 1. **Boot Loop:** The phone is stuck on the logo (e.g., the Samsung or Tecno logo) and will not turn on properly.
 2. **Forgotten Password:** When you are locked out of the phone and cannot reset it (flashing will wipe the password).
 3. **Virus/Malware:** To completely remove a deep-level virus.
 4. **"Bricked" Phone:** When a software update fails and the phone becomes "dead" or unresponsive.

5. Tools and Apps for Flashing Flashing is an advanced process and must be done carefully. Using the wrong file can "hard brick" (permanently kill) the phone.

1. **A Computer (PC):** Flashing is done *from* a computer.
2. **USB Cable:** To connect the phone to the computer.
3. **Flashing Software (on PC):**
 - **Odin:** For Samsung phones.
 - **SP Flash Tool:** For phones with MediaTek (MTK) chips (like many Tecno, Infinix).
 - **iTunes:** For Apple iPhones (this is called "Restoring").

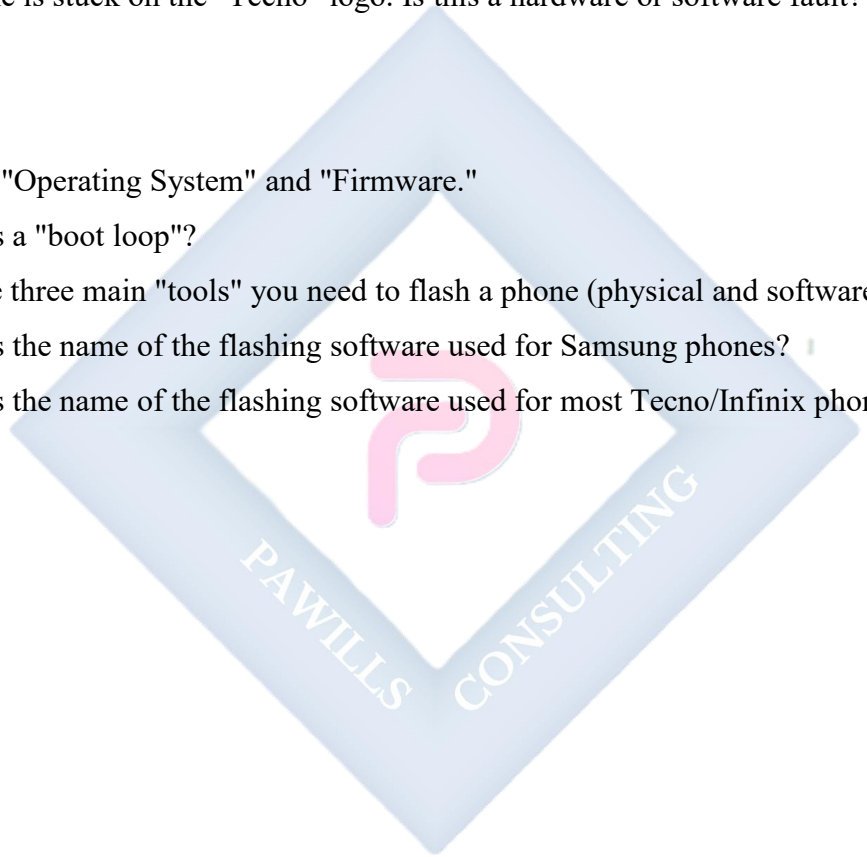
4. **The Correct Firmware File:** The *exact* OS file for your phone's *exact* model number.

Evaluation:

1. What is the difference between hardware and software?
2. What are the two main mobile operating systems?
3. What is "flashing"?
4. State two reasons why you should always update your phone's software.
5. A phone is stuck on the "Tecno" logo. Is this a hardware or software fault?

Assignment:

1. Define "Operating System" and "Firmware."
2. What is a "boot loop"?
3. List the three main "tools" you need to flash a phone (physical and software).
4. What is the name of the flashing software used for Samsung phones?
5. What is the name of the flashing software used for most Tecno/Infinix phones?



WEEK 9: FAULT DIAGNOSIS AND TESTING

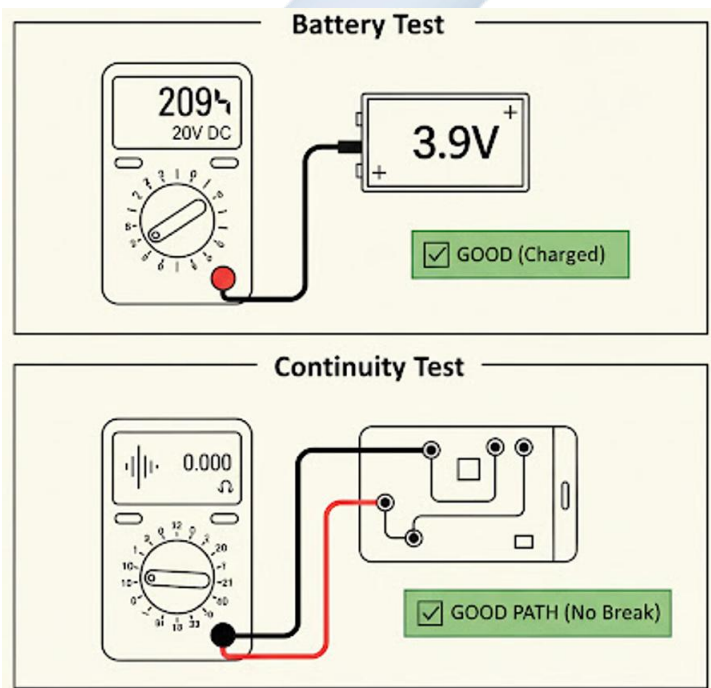
Topic: Fault Diagnosis and Testing

Content: Use of multimeter; Continuity test; Battery voltage test

Performance Objectives: By the end of this lesson, students should be able to:

1. Identify a "multimeter" as a diagnostic tool.
2. List the three main things a multimeter measures (Voltage, Current, Resistance).
3. Explain the purpose of the "Continuity Test."
4. Describe how to perform a battery voltage test.
5. Interpret the results of a battery voltage test.

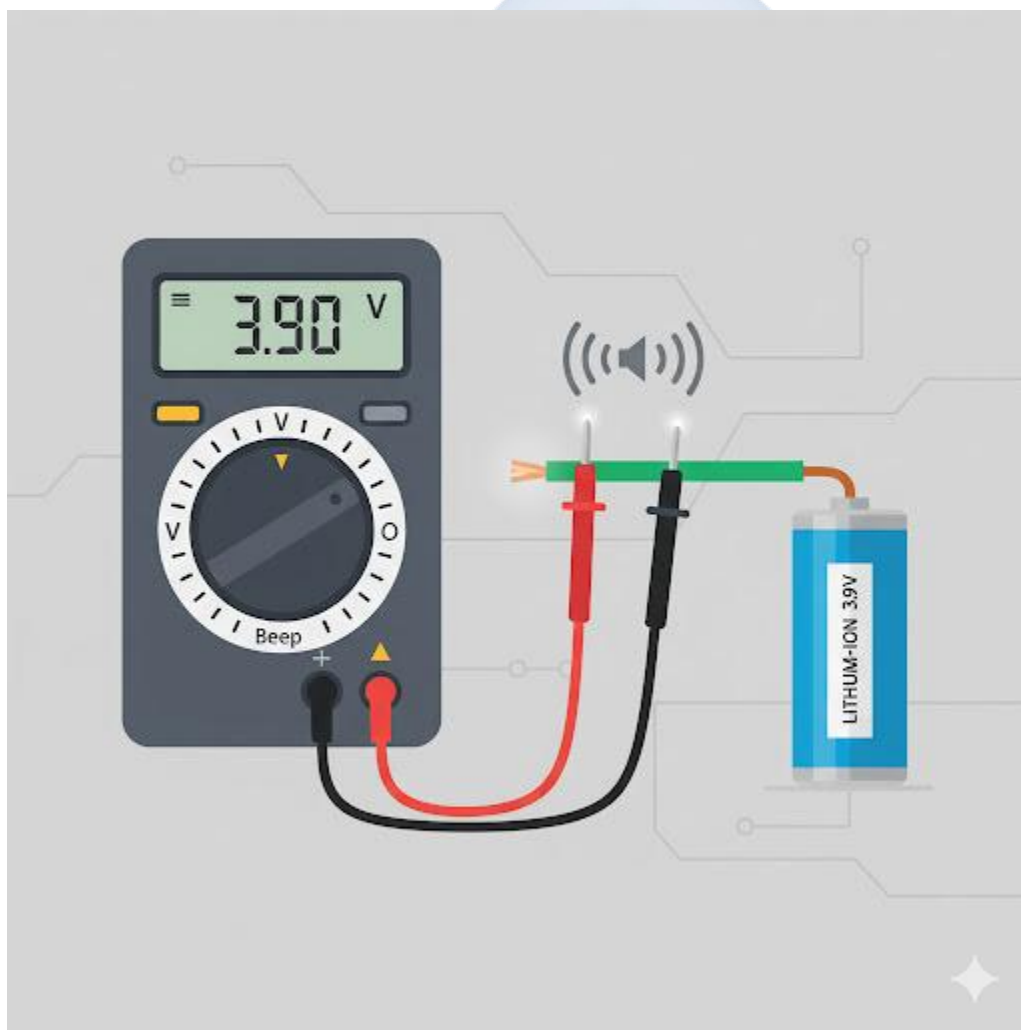
Lesson Content:



1. What is a Multimeter?

- **Definition:** A multimeter (or "multi-tester") is an electronic measuring instrument used by technicians to *diagnose* faults. It is our "doctor's stethoscope."

- **Function:** It can measure multiple electrical properties. For GSM repair, we use it for two main things:
 1. **DC Voltage (V):** To check if a battery is good.
 2. **Continuity (Beep Test):** To check if a path is broken.
- **Parts:**
 - **Display:** Shows the reading.
 - **Dial:** To select *what* you want to measure.
 - **Probes:** The red (positive) and black (negative) "needles" used to touch the circuit.



2. Test 1: Battery Voltage Test (Health Check)

- **Purpose:** To check if a battery is "dead" or "good."
- **Procedure:**
 1. Set the multimeter dial to **DC Voltage (V)**. (Use the 20V setting).
 2. Remove the battery from the phone (if possible).
 3. Identify the positive (+) and negative (-) terminals on the battery.
 4. Touch the **Red probe** to the **positive (+)** terminal.
 5. Touch the **Black probe** to the **negative (-)** terminal.
 6. Read the voltage on the display.
- **Interpreting the Results:**
 - **3.7V - 4.2V:** The battery is **GOOD** and healthy.
 - **3.2V - 3.6V:** The battery is **WEAK** (needs charging).
 - **0.0V - 3.0V:** The battery is **DEAD** or "deeply discharged." It may need to be "boosted" or replaced.

3. Test 2: Continuity Test (Beep Test)

- **Purpose:** To check if there is a complete, *unbroken* path between two points. This is used to find "breaks" in a circuit, like a faulty wire or a bad switch.
- **Procedure:**
 1. Set the multimeter dial to **Continuity Mode**. (It looks like a sound wave or a diode symbol:).
 2. Touch the Red and Black probes *together*. The multimeter should **BEEP**. This shows the tool is working.
 3. To test a component (e.g., a fuse, a switch, a wire), touch one probe to one end and the other probe to the other end.
- **Interpreting the Results:**
 - **If it BEEPS:** The path is **GOOD** (there is continuity).
 - **If it does NOT BEEP:** The path is **BROKEN** (it is an "open circuit"). The component is faulty and must be replaced.

- **Practical Use:** We use this to test power buttons, volume buttons, charging ports, and traces (paths) on the PCB.

Evaluation:

1. What is a multimeter?
2. What are the two main tests we use a multimeter for in phone repair?
3. What does the "Continuity Test" (Beep Test) check for?
4. If you do a continuity test and the multimeter *beeps*, is the path good or broken?
5. You test a battery and the multimeter reads 3.8V. Is the battery good or dead?

Assignment:

1. Draw a simple diagram of a multimeter and label its main parts (Display, Dial, Probes).
2. Describe the steps to perform a battery voltage test.
3. You test a battery and the multimeter reads 2.5V. What does this mean?
4. Describe the steps to perform a continuity test on a wire.
5. If you test a wire and the multimeter does *not* beep, what does this mean?

WEEK 10: PRACTICAL: FAULT DETECTION AND REPAIR

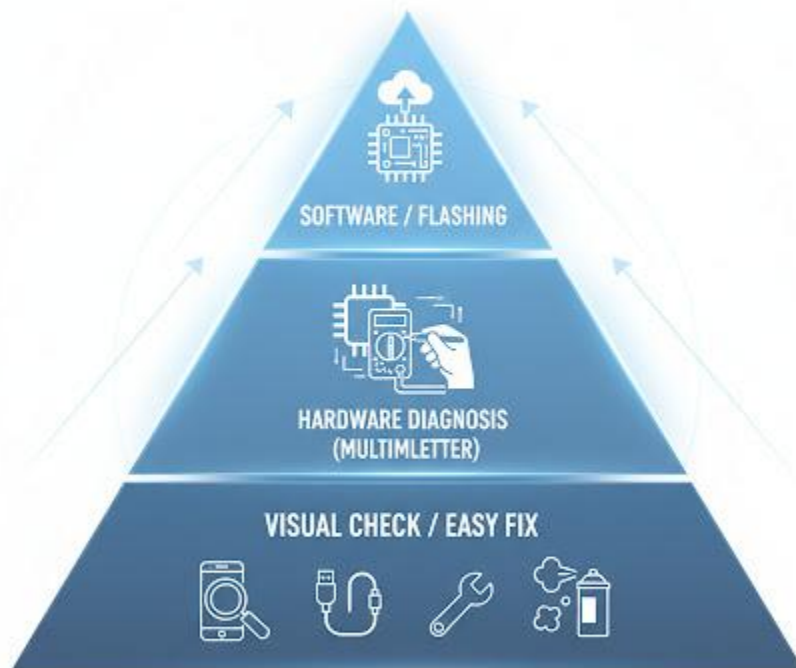
Topic: Practical: Fault Detection and Repair

Content: Step-by-step testing and replacement of faulty parts

Performance Objectives: By the end of this lesson, students should be able to:

1. List the three main stages of troubleshooting (Visual, Software, Hardware).
2. Follow a logical, step-by-step diagnostic process.
3. Apply the multimeter tests (from Week 9) in a practical scenario.
4. Practically identify a faulty charging port.
5. Safely disassemble and reassemble a simple phone (under supervision).

TROUBLESHOOTING STEPS



Lesson Content:

1. The Logic of Troubleshooting Troubleshooting is a step-by-step *process* of elimination. You must *never* guess. Always start with the *simplest* and *most common* solution first.

2. Scenario: A Customer brings a "Dead Phone" (Won't turn on)

- **Step 1: Gather Information (Talk to the client)**
 - *Question:* "What happened before it died?"
 - *Answer A:* "I dropped it in water." -> **Water Damage.**
 - *Answer B:* "I dropped it on the floor." -> **Physical Damage** (ICs or connectors may be loose).
 - *Answer C:* "Nothing, it just went off." -> **Battery or Software Fault.**
- **Step 2: Basic Visual & External Tests**
 1. Look for physical damage, water damage (check the indicator sticker).
 2. Plug in a known-good charger. Does a charging symbol appear?
 - *If YES:* The battery is just "deeply discharged." Let it charge for 30 mins.
 - *If NO:* Proceed to the next step.
- **Step 3: Internal Hardware Diagnosis (The Technician's Job)** (*This combines all lessons from the term*)
 0. **Disassemble:** Safely open the phone (using pry tools, screwdrivers).
 1. **Battery Test (Week 9):**
 - Test the battery with a multimeter.
 - *Result A:* Battery is 2.5V (Dead). → **Repair:** Boost or replace the battery.
 - *Result B:* Battery is 3.9V (Good). → The battery is not the problem. Proceed.
 2. **Visual Inspection (PCB):**
 - Look for corrosion (water damage), burned components, or loose connectors.
 3. **Continuity Test (Week 9):**
 - Test the Power Button. Set multimeter to "beep test."

- Press the button. Does it beep? If not, the power button is faulty. ->

Repair: Replace the power button.

4. Advanced Diagnosis:

- If the battery is good and the power button is good, the fault is likely in the **Power IC (Week 3)** or a **short circuit** on the PCB. This requires micro-soldering.

- **Step 4: Software Diagnosis (If the phone is stuck in a boot loop)**

0. If the phone turns on but is stuck on the logo...
1. Attempt a "Hard Reset" (Wipe data/factory reset) from recovery mode.
2. If that fails, the final step is to **"Flash" the firmware (Week 8)**.

Evaluation:

1. What is the *first* step in troubleshooting *any* fault? (Gather information / talk to the client).
2. What is the *second* step? (Visual inspection / check with a good charger).
3. You test a "dead" phone's battery, and it reads 3.9V. Is the battery the problem?
4. If the battery is 3.9V and the phone is dead, what is the *next* component you should test (using the continuity test)?
5. A phone is stuck on the Samsung logo. What is the final *software* solution for this?

Assignment:

1. What are the three main stages of troubleshooting? (Visual, Software, Hardware).
2. Why is it important to ask the client, "What happened?"
3. Create a step-by-step "Troubleshooting Guide" (like the one in the lesson) for a phone with **"No Network."**
 - (e.g., *Step 1: Check if the phone is in flight mode. Step 2: Try the SIM in another phone... etc.*)
4. Create a step-by-step "Troubleshooting Guide" for a phone where **"the other person can't hear me."**
 - (e.g., *Step 1: Check the microphone hole for dirt... etc.*)

5. (Safety) What is the main safety rule when opening a phone with a non-removable battery?



WEEK 11: REVISION

Topic: Revision

Content: Comprehensive review of all topics covered

Performance Objectives: By the end of this lesson, students should be able to:

1. Verbally define all key terms from the term (ESD, Circuit, Power IC, Flashing, Multimeter, Continuity).
2. Trace the power path for both charging and usage.
3. Differentiate between the earpiece, loudspeaker, and microphone.
4. Logically troubleshoot the 3 most common faults (No Power, No Charging, No Network).

Lesson Content:

1. Summary of Term 1 Topics

- **Week 1 (Review):** We remembered tools, parts, and **ESD Safety** (Static electricity is the silent killer of PCBs).
- **Week 2 (Circuits):** A circuit is a closed path. We learned the symbols (Resistor, Capacitor, Diode) and terms (Voltage, Current, Resistance).
- **Week 3 (Power Section):**
 - *Charging Path:* Port -> Charging IC -> Battery
 - *Power-On Path:* Battery -> **Power IC (PMIC)** -> CPU/Screen, etc.
- **Week 4 (Display/Network):**
 - *Display:* Screen (LCD/OLED) + **Digitizer** (Touch).
 - *Network:* SIM (Identity) + **Antenna** (Signal) + **Network IC** (Processor).
- **Week 6 (Sound/Charging):**
 - *Audio:* **Earpiece** (hear calls), **Loudspeaker** (ringer), **Microphone** (speak).
 - *Charging Port:* Most common fault. Always *clean* it first.
- **Week 8 (Software):** OS (Android/iOS). **Flashing** is re-installing the OS. Used for boot loops, forgotten passwords, or viruses.

- **Week 9 (Multimeter):** Our main test tool.
 - **Voltage Test:** Checks battery health (Good = 3.7V-4.2V).
 - **Continuity Test:** "Beep test" checks for broken paths.
- **Week 10 (Practical):** Always troubleshoot in order (Visual -> Battery Test -> Component Test).

2. Class Activity: Rapid-Fire Troubleshooting (Teacher gives a fault, students give the first 3 steps)

1. **Fault:** "Phone is not charging."
 - *Step 1:* Try new charger/cable.
 - *Step 2:* Visually inspect and clean the port.
 - *Step 3:* Test battery voltage / Replace charging port.
2. **Fault:** "Phone is stuck on the Tecno logo."
 - *Step 1:* Attempt a "hard reset" / factory reset.
 - *Step 2:* If it fails, "Flash" the phone.
 - *Step 3:* If flashing fails, it may be a hardware (e.g., memory) fault.
3. **Fault:** "Customer dropped the phone. Now the screen is black, but it rings."
 - *Step 1:* Open the phone.
 - *Step 2:* Check if the display flex cable is loose. Re-seat it.
 - *Step 3:* If it still fails, replace the screen assembly.

Evaluation: This week's evaluation is the First Term (Final) Examination.

Assignment:

1. Study all notes from the term.
2. Make a list of 10 key components (e.g., Power IC, Antenna) and their functions.
3. Make a list of 10 key terms (e.g., ESD, Flashing) and their definitions.
4. Write down the 3 most common faults and their *most likely* cause.
5. What is the most important safety rule you learned this term?

**SECOND TERM - JSS 2 COMPUTER HARDWARE AND GSM REPAIRS LESSON
NOTES**



WEEK 1: INTRODUCTION TO ENTREPRENEURSHIP MINDSET

Topic: Introduction to Entrepreneurship Mindset

Content: Meaning and importance; Thinking like an entrepreneur

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "entrepreneur" and "entrepreneurship."
2. Define "entrepreneurial mindset."
3. List at least five characteristics of an entrepreneurial mindset.
4. Explain the importance of entrepreneurship in the GSM repair field.
5. Differentiate between a "fixed mindset" and a "growth mindset."

Lesson Content:

1. Meaning of Entrepreneurship

- **Entrepreneur:** An individual who identifies a need or problem in the market, creates a business idea to solve it, and takes on the financial risks to start and manage that business, hoping to make a profit.
- **Entrepreneurship:** The *process* of designing, launching, and running a new business. It's not just about having a shop; it's about creating value.
- **Application to GSM:** A GSM repair technician is not just a person who fixes phones; they are an entrepreneur who provides a *service* (repair) to solve a *problem* (broken phone) for a *customer* (phone owner).

2. What is an "Entrepreneurial Mindset"?

- **Definition:** A specific way of thinking that allows you to see opportunities, take calculated risks, and learn from failure. It's about being a *problem-solver*, not just an employee.
- **Fixed Mindset vs. Growth Mindset:**

- **Fixed Mindset:** Believes skills are inborn. Avoids challenges. Gives up easily. (e.g., "I can't fix this new phone model; it's too hard.")
- **Growth Mindset:** Believes skills are built through effort. Embraces challenges. Learns from failure. (e.g., "I haven't fixed this model before. Let me watch a tutorial, read the manual, and learn how.")
- An entrepreneur *must* have a growth mindset.



3. Characteristics of an Entrepreneurial Mindset

1. **Problem-Solving:** You see a broken phone as a *puzzle* to be solved, not a frustration.
2. **Resilience / Persistence:** You don't give up when a repair is difficult. You keep trying new methods.

3. **Creativity / Innovation:** You think of new ways to do things (e.g., "How can I repair this faster?" "Can I offer a home-pickup service?").
4. **Risk-Taking:** You are willing to take *calculated* risks (e.g., spending money on a new, expensive tool, believing it will help you earn more in the future).
5. **Customer-Focused:** You are obsessed with making the customer happy, not just fixing the phone.
6. **Curiosity / Continuous Learning:** Technology changes fast. You must be willing to learn about new phones and new techniques *every day*.

4. Importance in GSM Repairs

- The GSM industry changes *very* fast. New phone models come out every few months.
- A technician with a *fixed mindset* will only know how to fix old phones and their business will die.
- A technician with an *entrepreneurial mindset* will see a new phone model as a new *opportunity* to learn and attract more customers. They will be the first to learn how to fix it.

Evaluation:

1. What is an "entrepreneur"?
2. What is an "entrepreneurial mindset"?
3. What is the difference between a fixed mindset and a growth mindset?
4. List three characteristics of an entrepreneur.
5. Why is a growth mindset so important for a GSM technician?

Assignment:

1. Define Entrepreneurship in your own words.
2. Explain the difference between an entrepreneur and someone with a "fixed mindset."
3. List five characteristics of an entrepreneurial mindset.
4. A technician tries to fix a screen but fails, and the customer is angry. How would someone with a *growth mindset* react?

5. (Personal) Think of a time you faced a difficult problem (like a hard math question). Did you use a fixed or growth mindset? Explain why.



WEEK 2: CONCEPT OF ENTREPRENEURSHIP

Topic: Concept of Entrepreneurship

Content: Identifying opportunities in GSM repair; Risk-taking and innovation

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "opportunity" in a business context.
2. Identify at least five different business opportunities within the GSM industry.
3. Define "risk" and list three types of risks in GSM repair.
4. Define "innovation" and provide two examples for a GSM business.

Lesson Content:

1. Identifying Opportunities in GSM Repair An "opportunity" is a chance to offer a product or service that people need or want, and are willing to pay for. The GSM industry is *full* of opportunities beyond just fixing a broken screen.

- **Core Opportunity:**

1. **Hardware Repairs:** This is the main one.

- Screen replacement
- Charging port replacement
- Battery replacement
- Speaker/Microphone repair
- Water damage cleaning

- **Other Opportunities (Specialization):**

2. **Software Services:**

- ❖ Flashing / Software updates (from Term 1)
- ❖ Unlocking (network or password)
- ❖ Virus and malware removal

3. Sales of Accessories:

- ❖ Screen protectors (tempered glass)
- ❖ Phone cases
- ❖ Chargers and cables
- ❖ Power banks
- ❖ Earphones

4. Refurbishing and Sales:

- ❖ Buying old or damaged phones cheaply.
- ❖ Repairing them (refurbishing).
- ❖ Selling them for a profit.

5. Micro-soldering (Advanced):

- ❖ Becoming a "technician's technician."
- ❖ Specializing in repairing the PCB (motherboard) itself (e.g., replacing ICs). This is a high-skill, high-profit area.

2. Understanding Risk-Taking

- **Risk:** The possibility of losing time, money, or reputation. Entrepreneurship is *always* risky.
- **Types of Risks in GSM Repair:**
 1. **Financial Risk:** You spend N50,000 on new tools and parts. What if no customers come? You have lost that money.
 2. **Technical Risk (Damaging the Phone):** While trying to fix a charging port, you accidentally damage the phone's screen. You now have to *buy* the customer a new screen, losing money and time.
 3. **Reputation Risk:** You do a bad repair. The customer tells all their friends that your shop is terrible. You lose future business.

- **Calculated Risk:** An entrepreneur *calculates* the risk. "This hot air station costs N20,000 (risk), but it will let me fix ICs and I can charge N5,000 per repair. I only need 4 repairs to make my money back." (calculated).



3. Understanding Innovation

- **Innovation:** This is *not* just "invention" (creating something new). It is *thinking of a new way to do something* to provide better value.
- **Examples of Innovation in a GSM Repair Business:**
 - **Service Innovation:**

- "Home/Office Service": You travel to the customer's house to fix their phone. (Saves them time).
- "1-Hour Screen Repair": Offering a *faster* service than your competitors.
- **Marketing Innovation:**
 - Using social media (Instagram, WhatsApp) to show "before and after" videos of your repairs.
- **Convenience Innovation:**
 - Sending a customer a text message when their repair is ready.
 - Offering a 30-day warranty on your repairs.

Evaluation:

1. What is a business "opportunity"?
2. List three different opportunities in the GSM industry (e.g., hardware, software, accessories).
3. What is "risk"?
4. Give one example of a "technical risk" in phone repair.
5. What is "innovation"? Give one example for a repair shop.

Assignment:

1. List five different services a GSM repair entrepreneur could offer.
2. What is the difference between a "calculated risk" and a "careless risk"?
3. A technician wants to start a repair business. What is the *financial risk*? What is the *reputation risk*?
4. Think about the repair shops near your house. What is one "innovation" you could introduce to make your business better than theirs?
5. What is "refurbishing"?

WEEK 3: CONCEPT OF MONEY

Use of money in business transactions; Savings and reinvestment

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "money."
2. List the three main functions of money.
3. Differentiate between "cash" and "cashless" (digital) forms of money.
4. Explain the role of money in a GSM repair business (transactions).
5. Define "savings" and "reinvestment" and explain their importance.

Lesson Content:

1. What is Money?

- **Definition:** Money is any item or verifiable record that is generally accepted as **payment** for goods and services. It is a tool that makes transactions easy.
- Before money, people used **Trade by Barter** (e.g., "I will fix your phone if you give me a chicken."). This was inefficient. Money solves this problem.



2. Functions of Money

Money has three main jobs:

1. **Medium of Exchange:** This is its most important function. It is the "middle-man" in a transaction. You accept money (Naira) for your repair service, and then you use that *same* money to buy food.
2. **Store of Value:** You can *save* money. It holds its value (or worth) over time, so you can earn it today and spend it next month.
3. **Unit of Account:** It is a "measuring stick." It allows us to put a *price* (a value) on things (e.g., "This screen repair is N5,000," "This charger is N1,500.").

3. Forms of Money in Agribusiness (and GSM Repair)

- **Cash:** Physical banknotes (Naira notes) and coins.
- **Cashless (Digital) Money:**
 - **Bank Transfers:** (e.g., using a bank app). This is the most common for technicians.
 - **POS Terminals:** Using a debit card.
 - **Online Payments:** (e.g., Paystack, Flutterwave).

4. The Role of Money in a GSM Repair Business

Money *flows* through the business.

- **Money In (Income):**
 - Payment from customers for repairs.
 - Sales of accessories (chargers, cases).
 - Sales of refurbished phones.
- **Money Out (Expenses):**
 - **Startup Costs:** Buying your first set of tools (screwdriver, multimeter, etc.).
 - **Running Costs (Overhead):**
 - Rent for the shop.
 - Electricity (NEPA bill) or fuel for the generator.
 - Data (to download firmware).

- **Cost of Goods Sold (COGS):**

- Buying new screens, batteries, and charging ports to use for repairs.

5. Savings and Reinvestment (The Key to Growth)

- **Profit:** The money left over after you pay all your expenses. (Profit = Income - Expenses).
- **Savings:** The portion of the profit you *keep* for emergencies or future use. (e.g., "What if my soldering iron breaks? I need savings to buy a new one.").
- **Reinvestment:** This is the *secret* to growing your business. It is the act of *using your profit to buy more assets* for the business.
 - **Example:** You make N50,000 profit.
 - **Bad Business:** You spend it all on personal items. Your business *never grows*.
 - **Good Business (Reinvestment):** You take N20,000 and *reinvest* it. You buy a new, advanced tool (like a hot air station) or buy 10 phone screens in bulk (at a discount).
 - Reinvestment allows you to *offer more services*, *work faster*, and *make more profit* in the future.

Evaluation:

1. What is "money"?
2. What is "trade by barter," and what is its main problem?
3. List the three main functions of money.
4. What is "reinvestment"?
5. Why is reinvestment more important for business growth than just saving?

Assignment:

1. Define Money, Savings, and Reinvestment.

2. A technician fixes a screen for N5,000 (Income). The screen part cost N3,000 (Expense).
What is the technician's profit?
3. List two examples of "Money In" (Income) for a GSM shop.
4. List three examples of "Money Out" (Expenses) for a GSM shop.
5. (Scenario) A technician makes N20,000 profit. Give one example of "Saving" it and one example of "Reinvesting" it.



WEEK 4: BOOKKEEPING IN GSM REPAIRS

Business records — daily sales, repairs log, expenses; Balance sheet basics

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "bookkeeping."
2. Identify and state the purpose of the three most important records for a technician.
3. Define "Asset," "Liability," and "Equity" (Capital).
4. Draft a simple Repairs Log.
5. Explain how bookkeeping helps in calculating profit.

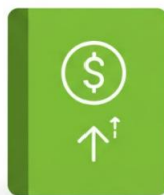
Lesson Content:

1. What is Bookkeeping?

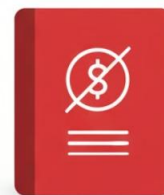
- **Definition:** Bookkeeping is the *organized process* of **recording** all the financial transactions of a business. It is the "diary" of every Naira that comes in and goes out.
- **Importance:**
 - It shows you if you are **making a profit** or a **loss**.
 - It helps you **track your expenses** (e.g., "Am I spending too much on data?").
 - It is needed for **paying taxes**.
 - It helps you **plan for the future** (e.g., "I have saved N50,000; I can now buy that new tool.").



Repairs Log



Income



Expenses

2. Key Records for a GSM Repair Business You don't need complex software. You can start with three simple notebooks (ledgers).

1. Repairs Log / Job Log:

- **Purpose:** To track every phone that comes into your shop. This is crucial for organization and customer service.

Date In	Customer Name	Phone No.	Phone Model	Fault Description	Quoted Price (₦)	Date Out
Oct 28	Mr. John	080...	Tecno Pova	Broken Screen	8,000	Oct 29
Oct 28	Bisi	070...	iPhone X	Won't charge	5,000	Oct 28

2. Daily Sales / Income Ledger:

- **Purpose:** To record all money *received*.

Date	Description of Service / Item	Amount Received (₦)
Oct 29	Repair: Tecno Pova (Mr. John)	8,000
Oct 29	Sale: 1× iPhone Charger	1,500
	Total for the Day:	9,500

3. Expenses / Expenditure Ledger:

- **Purpose:** To record all money *spent*.

Date	Description of Expense	Amount Spent (₦)
Oct 29	Buy 1× Tecno Pova Screen (for Mr. John)	5,000
Oct 29	Buy 10× iPhone Chargers (Stock)	8,000
Oct 29	Airtime / Data	1,000
	Total for the Day:	14,000

3. Calculating Profit

At the end of the week or month, you calculate your profit: **Profit = Total Income - Total Expenses**

- *Example:*
 - Total Income (from Sales Ledger) = N150,000
 - Total Expenses (from Expenses Ledger) = N90,000
 - **Profit** = N150,000 - N90,000 = **N60,000**

4. Balance Sheet Basics (Simple Version)

A balance sheet shows the *health* of your business at one point in time. It follows one rule: **Assets = Liabilities + Equity**

- **Assets:** Everything the business *owns* that has value.
 - Cash in the bank (N50,000)
 - Your tools (N30,000)
 - Your parts inventory (10 screens worth N40,000)
 - **Total Assets = N120,000**
- **Liabilities:** Everything the business *owes* to others.
 - A loan from your friend (N20,000)
- **Equity (or Capital):** The *owner's* investment in the business. This is what's left over.
 - Equity = Assets - Liabilities
 - Equity = N120,000 - N20,000 = N100,000

Evaluation:

1. What is "bookkeeping"?
2. Why is a "Repairs Log" important?
3. What is the difference between an Income Ledger and an Expenses Ledger?
4. What is the formula for calculating Profit?
5. What are "Assets"? Give one example for a GSM shop.

Assignment:

1. Define Bookkeeping, Asset, and Liability.

2. (Practical) Draw a simple "Repairs Log" with at least 5 columns.
3. (Practical) Draw a simple "Expenses Ledger" with 3 columns.
4. A technician makes N10,000 from two screen repairs. This is (Income / Expense).
5. He buys a new charging port for N1,500. This is (Income / Expense).
6. At the end of the month, his Total Income is N80,000 and his Total Expenses are N55,000. What is his profit?



WEEK 6: BUSINESS STARTUP IN GSM REPAIRS

Writing a simple business plan; Budgeting and equipment listing

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "business plan."
2. List the three main sections of a simple business plan.
3. Define "budget" and "startup costs."
4. Create a basic equipment list for a new GSM repair business.
5. Draft a one-page simple business plan.

Lesson Content:

1. What is a Business Plan?

- **Definition:** A business plan is a *written document* that describes your business, its goals, and how you plan to achieve those goals.
- **Purpose:** It is your "map" or "blueprint" for success. It helps you think clearly about your business and can be shown to a bank or investor to get a loan.



ITEM	DESCRIPTION	EST. COST (USD)
	Precision Screwdriver Kit	\$45.99
	Digital Multimeter	\$89.50
	New Screens & Components	\$450.00
TOTAL ESTIMATED COST:		USD \$485.49

2. Sections of a Simple Business Plan (for a JSS2 Student) A simple plan answers four questions: *What, Who, Where, and How Much?*

- **Section 1: The Business Idea (What & Who)**
 - **Business Name:** (e.g., "Deji's Phone Clinic," "FastFix Repairs")
 - **Owner:** (Your name)
 - **Business Goal (Mission):** (e.g., "To provide fast, honest, and reliable phone repairs for students and families in my area.")
 - **Services (What will I sell?):** (e.g., Screen repair, Battery replacement, Software flashing, Selling accessories).
- **Section 2: The Market (Where & Who)**
 - **Location:** (e.g., "A small kiosk at my street junction" or "A mobile service from my home.")
 - **Target Customers (Who will buy?):** (e.g., "Students in my school," "People in my neighbourhood," "Other repair shops.")
 - **Competitors (Who else is doing it?):** (e.g., "The big shop on the main road. My advantage will be lower price and faster service.")
- **Section 3: The Money (How Much?)**
 - This is the **Budget or Financial Plan.**

3. Budgeting and Equipment Listing (Startup Costs)

- **Startup Costs:** The *total amount of money* you need to *start* the business (before you make your first sale).
- **Budget:** A list of all your expected startup costs.

Example: Basic Startup Budget & Equipment List

Item / Category	Description	Estimated Cost (N)
1. Tools (Equipment)		
	Precision Screwdriver Kit	3,000

	Plastic Pry Tools & Picks	1,000
	ESD-Safe Tweezers	500
	Digital Multimeter	4,000
	Soldering Iron (basic)	2,500
	Subtotal (Tools):	N11,000
2. Parts (Inventory)		
	2x Tecno replacement screens	8,000
	2x Infinix charging ports	1,000
	5x iPhone chargers (for sale)	4,000
	Subtotal (Parts):	N13,000
3. Other Costs		
	Business Cards / Fliers	2,000
	First month's rent (if in a shop)	10,000
	Subtotal (Other):	N12,000
TOTAL STARTUP COST:		N36,000

Evaluation:

1. What is a "business plan"?
2. What is its main purpose? (It's a "map").
3. What are "startup costs"?
4. List three main categories in a simple budget.
5. List five *essential* tools for a startup equipment list.

Assignment:

1. Define "Business Plan" and "Budget."
2. (Practical) Create a name and a one-sentence "Business Goal" for your own future repair shop.
3. (Practical) List all the services you will offer in your shop (at least 3).
4. (Practical) List who your "target customers" will be.

5. (Practical) Using the example, create a simple budget. You have N15,000 to start. List the tools and parts you would buy.



WEEK 8: CUSTOMER RELATIONS

Content: Handling customers professionally; Honesty and reliability in service delivery

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "customer relations" (or "customer service").
2. Explain why good customer service is vital for a repair business (trust, repeat business).
3. List five principles of professional customer handling.
4. Define "honesty" and "reliability" in the context of repairs.
5. Describe the steps for handling an angry or dissatisfied customer.

Lesson Content:

1. What is Customer Relations?

- **Definition:** Customer relations (or customer service) is the way a business *communicates and interacts* with its customers. It is the *experience* you give the customer, not just the repair.
- **Why it's VITAL:**
 - In the repair business, the customer is already *stressed* (their phone is broken). Your job is to be the *calm, professional solution*.
 - A good repair + bad service = **No repeat customer.**
 - A good repair + good service = **A loyal customer for life** (and 5 of their friends).
 - Trust is everything. The customer is trusting you with their *private* device (photos, messages).

2. Principles of Professional Customer Handling

1. **First Impression (The Greeting):** Greet the customer politely and with a smile. (e.g., "Good morning, welcome. How can I help you?").
2. **Active Listening (The Diagnosis):**
 - Stop what you are doing. Give them your full attention.
 - Listen *carefully* to their problem (e.g., "I dropped it in water...").

- Ask questions ("When did this happen?" "Have you tried anything?").

3. Clear Communication (The Quote):

- Be honest and clear. *Do not* use technical jargon.
- *Bad*: "The PMIC has a shorted VCC_Main."
- *Good*: "The main power chip on the board is damaged. I will need to replace it. It will cost N4,000."
- *Always* give the price *before* you start the work.

4. Manage Expectations (Reliability):

- **Reliability**: Doing what you promised to do.
- *Do not* over-promise. (e.g., "It will be ready in 1 hour!") if you know it will take 3 hours.
- *Better*: "I need 3 hours to fix this properly. I will send you a text message as soon as it is ready."

5. Professional Appearance: Your shop must be clean, and you must look presentable.

3. Honesty and Reliability

• Honesty:

- *Never* lie about a fault. If the fault is just a dirty charging port, *clean it* and charge a small fee. Do not lie and say you replaced the port.
- *Never* steal parts. Do not take the customer's original battery and replace it with an old one.
- *Protect their data*. Never look at a customer's photos or messages.

• Reliability:

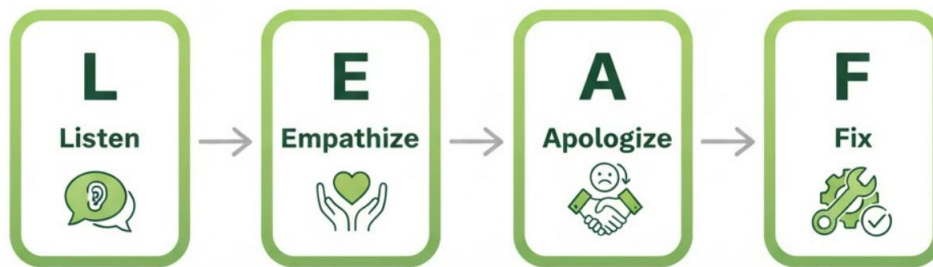
- If you say the phone will be ready on Tuesday, it *must* be ready on Tuesday.
- If you give a 30-day warranty, you must *honour* it if the part fails.

4. Handling Complaints (The "L.E.A.F." Method) When a customer is angry (e.g., "You fixed my screen, and now my speaker is not working!"), *do not* argue.

1. L - Listen: Let them vent their anger. Do not interrupt.

2. **E - Empathize:** Show you understand their frustration. (e.g., "I understand, that is very frustrating. I would be upset too.").
3. **A - Apologize:** Apologize for the *experience*. (e.g., "I am very sorry for this inconvenience.").
4. **F - Fix:** Offer a clear solution. (e.g., "Let me open the phone right now and check the speaker connection. It may have come loose. I will fix it for free.").

The L.E.A.F. Complaint Flow



Evaluation:

1. What is "customer relations"?
2. Why is "trust" so important in the GSM repair business?
3. What does it mean to "manage expectations"?
4. Give one example of *dishonesty* in GSM repair.
5. What are the four steps (L.E.A.F.) for handling an angry customer?

Assignment:

1. Define Customer Relations, Honesty, and Reliability.
2. Why is it better to tell a customer a repair will take 3 hours (and finish in 2) than to say it will take 1 hour (and finish in 2)?
3. A customer gives you their phone to fix the screen. You notice their private messages popping up. What is the *ethical* and *professional* thing to do?

4. (Role-play) A customer is angry because their new screen stopped working. Write down the *exact words* you would say for "Listen," "Empathize," "Apologize," and "Fix."
5. List 5 qualities of a professional GSM technician.



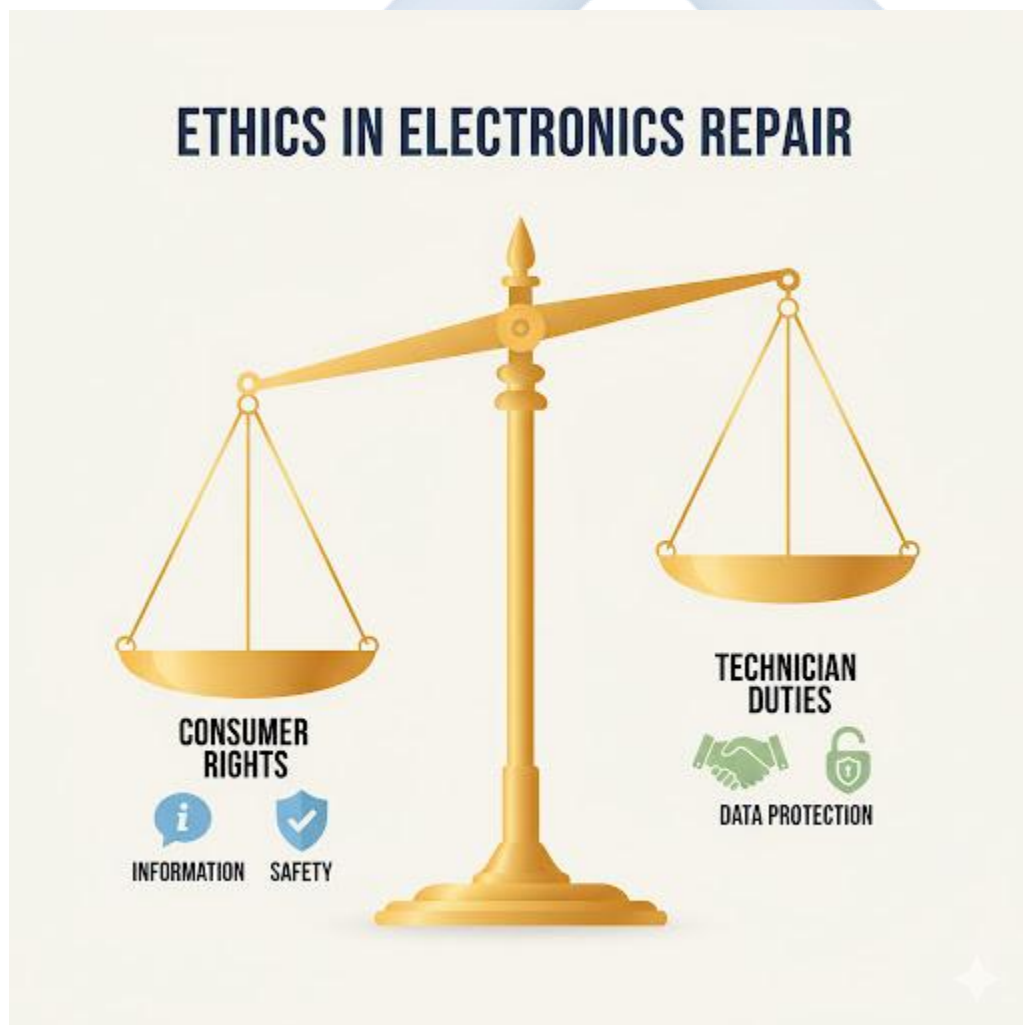
WEEK 9: RIGHTS AND DUTIES OF CITIZENS IN BUSINESS

Topic: Rights and Duties of Citizens in Business

Content: Consumer rights and seller responsibilities; Ethical practices in mobile repair

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "consumer" and "consumer rights."
2. List three rights of a consumer (customer).
3. List three duties (responsibilities) of a seller (technician).
4. Define "ethics" in a business context.
5. Identify two unethical practices in GSM repair.



Lesson Content:

1. Who is a Consumer?

- **Consumer:** A person who *pays* for goods (like a charger) or *services* (like a repair). In our business, the *customer* is the consumer.
- **Government Agencies:** In Nigeria, agencies like the **FCCPC (Federal Competition and Consumer Protection Commission)** and **SON (Standards Organisation of Nigeria)** exist to protect consumers.

2. Key Rights of the Consumer (The Customer) A customer has many rights, including:

1. **The Right to Safety:** The right to be protected from dangerous goods or services. (e.g., a customer has the right to a repair that does not make their battery explode).
2. **The Right to Information:** The right to be given the *facts* needed to make an informed choice. (e.g., "Is this an original screen or a copy?" "How much will it cost?").
3. **The Right to Choose:** The right to select from a range of products and services at fair prices.
4. **The Right to be Heard (Redress):** The right to *complain* and receive compensation (a refund, a free re-repair) if the service was bad.

3. Duties and Responsibilities of the Seller (The Technician) As technicians, we have a matching *duty* (responsibility) for every right the consumer has.

1. **Duty to Provide Safe Service:** We must use safe parts and follow safety procedures (like ESD rules) to protect the customer's device.
2. **Duty to Provide Honest Information:** We *must* be honest about the fault, the price, and the parts we are using.
3. **Duty to Protect Customer Data:** We have a legal and ethical duty to protect the *privacy* of the customer's photos, messages, and contacts. (This is a form of "Right to Safety").
4. **Duty to Provide Redress:** We must handle complaints professionally (Week 8) and, if we are at fault, we must fix the problem.

4. Ethical Practices vs. Unethical Practices

- **Ethics:** The moral rules of conduct. "What is the *right* thing to do?"
- **Ethical Practices (Good):**
 - Being honest about the problem (e.g., "Your charging port is just dirty, I won't charge you to clean it.").
 - Using the *exact* part you said you would use (e.g., if the customer paid for an *original* screen, you must use an original screen).
 - Giving a fair price.
 - Protecting all customer data.
 - Warning the customer of any risks (e.g., "Fixing this water damage is 50/50. It may not work.").
- **Unethical Practices (Bad):**
 - **Parts Swapping:** Stealing a customer's good, original component (like a battery or screen) and replacing it with a fake or old one.
 - **Fake Diagnosis:** Lying about the fault to charge more money. (e.g., The phone just needs a new battery, but you tell the customer the "Power IC is bad" and charge them 5x the price).
 - **Snooping:** Looking through a customer's private photos, messages, or contacts. This is a *major* ethical violation.
 - Using low-quality, "copy" parts and charging the price for "original" parts.

Evaluation:

1. Who is a "consumer"?
2. What is the "Right to Information"?
3. What is a seller's "Duty to Provide Redress"?
4. What is "parts swapping," and why is it unethical?
5. Give one other example of an *unethical* practice in GSM repair.

Assignment:

1. Define "Consumer Rights" and "Business Ethics."

2. List three rights of a consumer.
3. List three duties of a GSM repair technician.
4. A customer gives you their phone and password to test the speaker. What is the *ethical* and *unethical* way to handle their private data?
5. Why is it important to be honest about using "original" vs. "copy" (fake) parts?



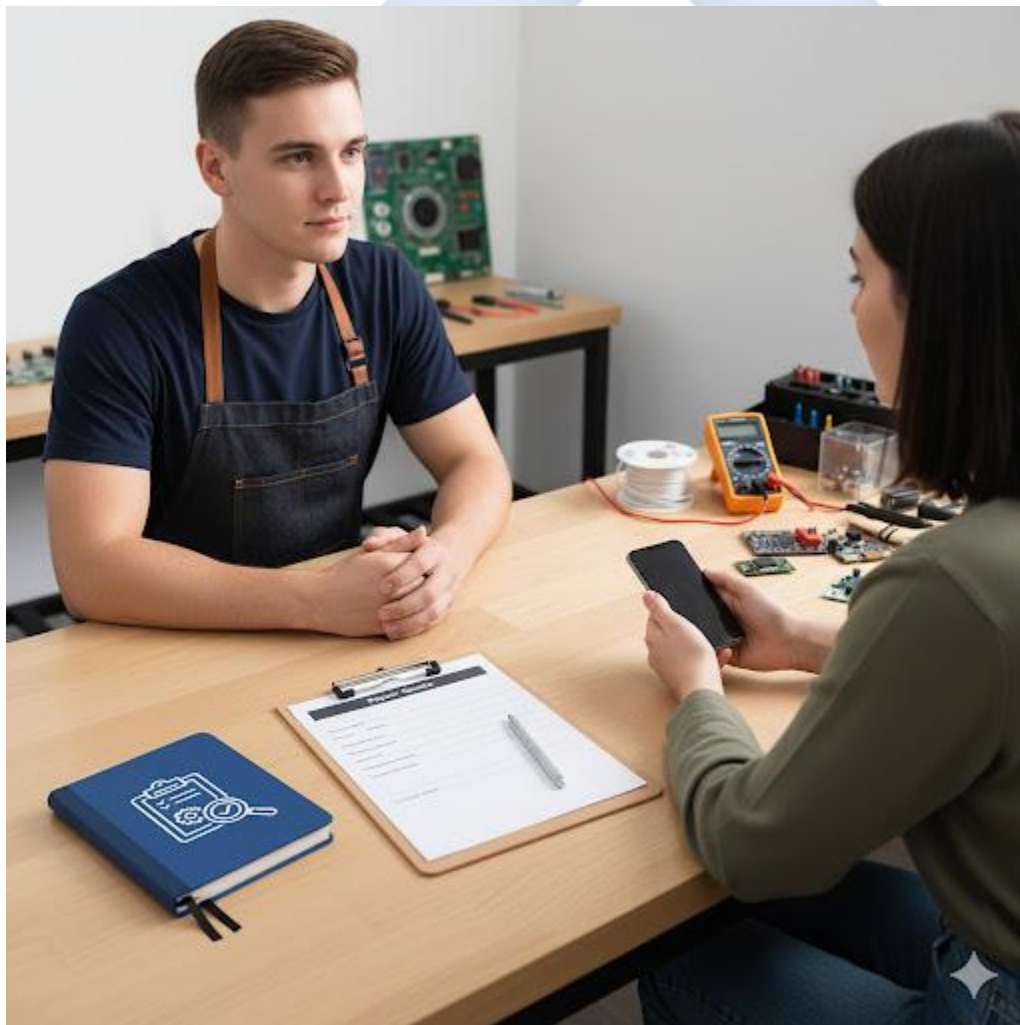
WEEK 10: PRACTICAL: BUSINESS SIMULATION

Topic: Practical: Business Simulation

Content: Role-play of shop management and customer service

Performance Objectives: By the end of this lesson, students should be able to:

1. Demonstrate the correct greeting and diagnostic procedure.
2. Demonstrate how to give a clear and honest price quote.
3. Demonstrate how to manage customer expectations (time).
4. Demonstrate the "L.E.A.F." method for handling a complaint.
5. Practice the basics of bookkeeping (filling out a repair log).



Lesson Content:

1. Activity Setup

- **Goal:** To combine all the lessons of the term (Entrepreneurship, Bookkeeping, Customer Service, Ethics) into one practical activity.
- **Setup:** Divide the class into groups of 3-4.
 - **Role 1: The Technician / Entrepreneur** (The "Boss")
 - **Role 2: The Customer A** (with a simple fault)
 - **Role 3: The Customer B** (an angry customer with a complaint)
 - **Role 4: The Bookkeeper** (sits quietly and fills in the records)
- **Materials:** A simple "Repairs Log" sheet (from Week 4), a "Sales" sheet, a "Props" desk (with a few old phones, chargers, tools).

2. Simulation Scenarios

- **Scenario 1: The New Customer**
 1. **Customer A** walks into the "shop."
 2. **Technician** must:
 - Greet them politely (e.g., "Welcome, how can I help?").
 - Practice *Active Listening* (Customer: "My phone won't charge.").
 - Ask diagnostic questions (e.g., "Did it fall? Have you tried another charger?").
 - Perform a quick *visual diagnosis* (e.g., "Let me look at the port. Ah, it's full of dirt.").
 - Give a clear *quote and managing expectations* (e.g., "It looks like the port is just dirty. I can clean it for N500. It will take 10 minutes.").
 3. **Bookkeeper** must:
 - Fill in the **Repairs Log** (Customer name, phone, fault, price).
 4. **Technician** (simulates the repair) and hands the phone back.
 5. **Customer A** pays.
 6. **Bookkeeper** must:

- Fill in the **Sales Ledger** (e.g., "Port Cleaning - N500").
- **Scenario 2: The Angry Customer**
 1. **Customer B** storms into the shop. ("I was here yesterday! You fixed my screen, and now my speaker is not working! This is rubbish!").
 2. **Technician** must use the **L.E.A.F. Method (Week 8)**:
 - **Listen:** (Let them finish. Don't interrupt. Nod.)
 - **Empathize:** ("I understand, that is very frustrating. You paid for a repair, and now have a new problem.")
 - **Apologize:** ("I am very sorry for this inconvenience. Let's look at it immediately.")
 - **Fix:** ("It is possible the speaker connection came loose during the repair. Please give me 15 minutes, and I will open it and fix it for you right now, for free, of course.")
 3. **Technician** (simulates the repair) and hands it back.
 4. **Customer B** (should be calm now) says thank you.
 5. **Bookkeeper** must:
 - Make a note in the *Repairs Log* (e.g., "Customer returned, speaker issue. Fixed for free (warranty).")

Evaluation: (Observation-based)

1. Did the technician greet the customer professionally?
2. Did the technician give a clear quote *before* the work?
3. Did the bookkeeper fill in the Repairs Log correctly?
4. Did the technician *argue* with the angry customer, or did they use the L.E.A.F. method?
5. Did the technician show honesty in their diagnosis?

Assignment:

1. Why is a "Business Simulation" (role-play) a useful way to learn?
2. Write two sentences a *good* technician should say when greeting a new customer.
3. Write two sentences a *bad* technician might say when handling an angry customer.

4. Write the two sentences a *good* technician should say.
5. What is the most important lesson you learned from the business simulation?



WEEK 11: REVISION

Topic: Revision

Content: Comprehensive review of all topics covered

Performance Objectives: By the end of this lesson, students should be able to:

1. Verbally define all key terms from the term (Entrepreneurship, Risk, Innovation, Bookkeeping, Profit, Asset, Customer Relations, Ethics).
2. Explain the importance of a growth mindset.
3. List the key records a technician must keep.
4. Explain the "L.E.A.F." method for complaint handling.
5. List three consumer rights and three technician duties.

Lesson Content:

1. Summary of Term 2 Topics (The "Business" Term) This entire term was about moving from being just a "technician" (someone who fixes things) to being an "entrepreneur" (someone who builds a business).

- **Week 1 & 2 (The Mindset):**
 - An **Entrepreneur** is a problem-solver who takes calculated **risks** and **innovates** (finds new ways to serve).
 - You must have a **Growth Mindset** (believing you can learn) not a *Fixed Mindset* (believing skills are fixed).
 - Opportunities in GSM are huge: Repairs, Software, Accessories, Sales.
- **Week 3 & 4 (The Money - Bookkeeping):**
 - **Money** is a tool for transactions.
 - **Reinvestment** (using profit to buy more assets) is the key to growth.
 - **Bookkeeping** is tracking all transactions.
 - **Key Records:** 1. Repairs Log, 2. Income Ledger, 3. Expenses Ledger.
 - **Profit = Total Income - Total Expenses**

- **Assets** (what you own) = **Liabilities** (what you owe) + **Equity** (your worth).
- **Week 6 (The Plan):**
 - A **Business Plan** is your "map" (Idea, Market, Money).
 - A **Budget** lists your **Startup Costs** (Tools, Parts, Rent).
- **Week 8 & 9 (The People - Ethics & Service):**
 - **Customer Service:** Your most important marketing tool. Trust is key.
 - **Honesty & Reliability:** Don't lie, don't steal parts, keep your promises.
 - **Complaints (L.E.A.F.):** Listen, Empathize, Apologize, Fix.
 - **Ethics:** Don't be "unethical" (e.g., parts swapping, fake diagnosis, snooping).
 - **Consumer Rights:** Customers have the right to safety, information, and redress.

2. Class Activity: Final Q&A (Teacher asks rapid-fire questions from the term to prepare for the final examination).

- "What is a growth mindset?"
- "What is the formula for profit?"
- "What is 'reinvestment'?"
- "What are the 3 key records?"
- "What does 'L.E.A.F.' stand for?"
- "What is 'parts swapping'?"

Evaluation: This week's evaluation is the Second Term (Final) Examination.

Assignment:

1. Study all notes from the term.
2. Make a list of 10 key "business" terms and their definitions.
3. Make a list of 10 "technical" terms from last term (e.g., Power IC, Flashing).
4. Explain why "Customer Service" is more important than "Low Prices" for building a long-term business.
5. What is the biggest difference between being an *employee* and being an *entrepreneur*?

**THIRD TERM - JSS 2 COMPUTER HARDWARE AND GSM REPAIRS LESSON
NOTES**



WEEK 1: INTRODUCTION TO SOFTWARE TOOLS

Topic: Introduction to Software Tools

Content: Common repair tools (Miracle Box, SP Flash Tool, etc.); Installation and usage

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Software Tools" in the context of GSM repair.
2. Differentiate between a "box/dongle" (hardware) and "flashing software" (software).
3. Identify at least three common software tools and their primary use.
4. Describe the general process for installing a software tool, including drivers.
5. State the importance of using the correct tool for the phone's chipset.



Lesson Content:

1. What are Software Tools in GSM Repair? (Recap from Term 1, Week 8)

- **Hardware** is the physical phone. **Software** is the Operating System (OS) and apps.
- **Software Tools** are specialized computer programs that allow a technician to communicate with a phone's internal memory (eMMC) and Central Processing Unit (CPU).
- **Purpose:** They are used to fix *software-level* problems, not hardware problems. Their main jobs are:
 - **Flashing:** Installing, re-installing, or upgrading the phone's firmware (OS).
 - **Unlocking:** Removing network locks (SIM lock) or security locks (passwords, patterns).
 - **Repairing:** Fixing software-corrupted data, like an unknown baseband or lost IMEI number.

2. Types of Software Tools

- **A. PC-Based Software (Free Tools):**
 - These are programs that run on your computer. They are often "leaked" or official tools from the chip manufacturers.
 - **SP Flash Tool (SmartPhone Flash Tool):**
 - **Use:** The *most common* tool for flashing phones with **MediaTek (MTK)** CPUs (e.g., most Tecno, Infinix, Gionee, and many other brands).
 - **How it works:** It communicates directly with the phone's CPU to load the firmware files.
 - **Odin:**
 - **Use:** The official *internal* tool for **Samsung** phones. It is used to flash official Samsung firmware.
 - **QFIL (Qualcomm Flash Image Loader):**
 - **Use:** A tool for flashing phones with **Qualcomm** CPUs.
- **B. Boxes and Dongles (Paid Tools):**

- These are small *hardware devices* (a box or a USB-like "dongle") that you plug into your computer.
- They run their own special software and are much more powerful than the free tools. They are "all-in-one" solutions for a professional technician.
- **Miracle Box / Miracle Thunder Dongle:**
 - **Use:** A very popular "all-in-one" tool. It supports a huge range of CPUs (MTK, Qualcomm, Spreadtrum).
 - **Functions:** Flashing, unlocking, formatting, repairing IMEI, and bypassing Factory Reset Protection (FRP).
- **Z3X Box / Octopus Box / UMT Dongle:**
 - **Use:** Other professional boxes, often specializing in certain brands like Samsung or LG, but also supporting many others.

3. Installation and Usage Installing these tools is a multi-step process:

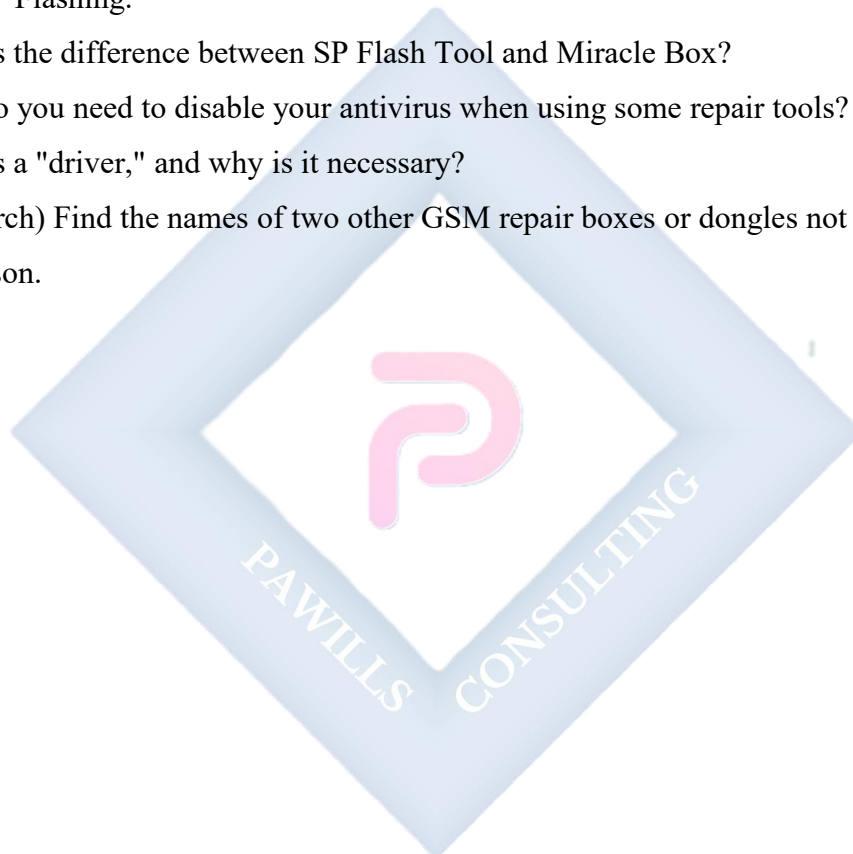
1. **Install Drivers:** This is the *most important* step. A "driver" is a small piece of software that teaches your computer how to "talk" to the specific phone CPU (e.g., you must install the "MTK VCOM Driver" before SP Flash Tool will work).
2. **Disable Antivirus (Temporarily):** Many professional tools (like Miracle Box) are falsely flagged as "viruses" because they use advanced methods to communicate with phones. You must disable your antivirus during installation and use.
3. **Install the Tool:** Run the .exe (executable) file to install the main program (e.g., flashtool.exe).
4. **Load Firmware:** Open the tool, click "Scatter-loading" (in SP Flash Tool) or "Load File" and select the firmware file (.pac, .md5) you want to flash.
5. **Connect the Phone:** Connect the phone (usually *switched off*) to the computer via USB.
6. **Start the Process:** Click "Download" (in SP Flash Tool) or "Start" (in Odin) to begin flashing.

Evaluation:

1. What are "software tools" used for in GSM repair? (List 2 uses).
2. What is the difference between a "dongle" and a "software-only" tool?
3. What does "SP Flash Tool" stand for, and what CPU type is it used for?
4. What is the *most important* step before installing a flashing tool? (Installing drivers).
5. What is the name of the flashing tool used for Samsung phones?

Assignment:

1. Define "Flashing."
2. What is the difference between SP Flash Tool and Miracle Box?
3. Why do you need to disable your antivirus when using some repair tools?
4. What is a "driver," and why is it necessary?
5. (Research) Find the names of two other GSM repair boxes or dongles not mentioned in the lesson.



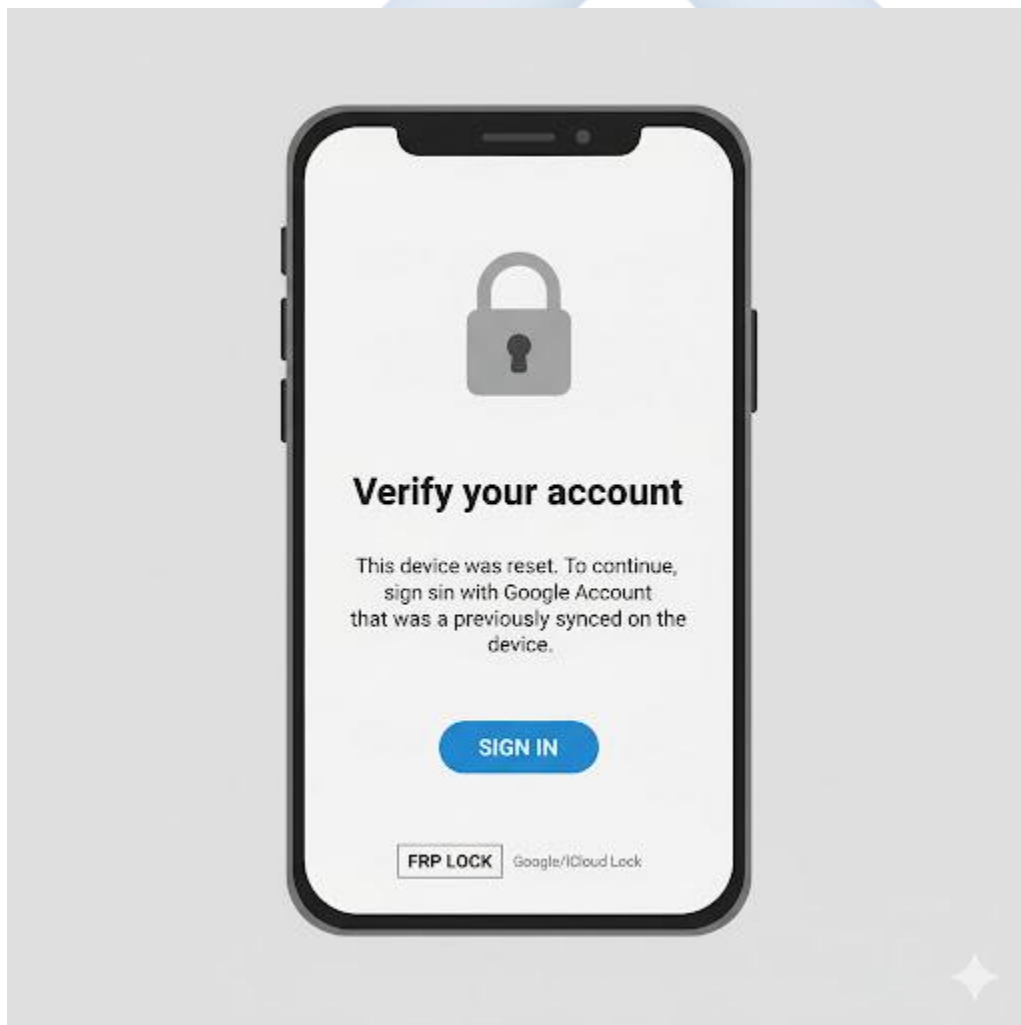
WEEK 2: PHONE UNLOCKING AND PASSWORD RESET

Topic: Phone Unlocking and Password Reset

Content: Security locks, factory reset, bypassing screen locks

Performance Objectives: By the end of this lesson, students should be able to:

1. List four types of security locks on a mobile phone.
2. Differentiate between a "SIM lock" and a "screen lock."
3. Describe the "Factory Reset" method for removing a screen lock.
4. Define "FRP (Factory Reset Protection)" and its purpose.
5. Explain how software tools are used to bypass screen locks.



Lesson Content:

1. Types of Security Locks

Security locks are used to protect a user's private data.

1. **Screen Lock:** Prevents unauthorized access to the phone's home screen.
 - **Pattern:** Draw a shape on a 3x3 grid.
 - **PIN (Personal Identification Number):** A numeric code (usually 4-6 digits).
 - **Password:** An alphanumeric code (letters, numbers, symbols).
 - **Biometrics:** Using your body (e.g., Fingerprint, Face ID).
2. **SIM Lock (Network Lock):**
 - This is a lock put on the phone by a *network provider* (e.g., AT&T in the USA) to stop you from using another provider's SIM card.
 - This is *not* the same as a "SIM PIN," which locks the card itself.
3. **FRP (Factory Reset Protection):**
 - A security feature by Google (for Android) and Apple (iCloud Lock).
 - If a phone is "factory reset" from recovery mode, it will *require* the user to enter the *last Google/Apple account* that was on the phone.
 - **Purpose:** To make a stolen phone *useless* to a thief. Even if they wipe it, they can't use it without the original owner's password.

2. Removing Screen Locks (PIN, Pattern, Password) Warning:

All these methods will **ERASE ALL USER DATA** (photos, contacts, messages).

- **Method 1: Factory Reset (from Recovery Mode)**

1. This is the standard, built-in method.
2. Turn off the phone.
3. Boot into "Recovery Mode" (e.g., by holding **Power + Volume Up**).
4. A menu appears. Use the volume buttons to navigate to **"Wipe data/factory reset"**.
5. Press the power button to select.
6. Confirm "Yes."
7. The phone is erased. Select **"Reboot system now."**

- **Problem:** If the user has FRP on, the phone will now be stuck on the Google/Apple account login screen.
- **Method 2: Bypassing/Resetting with Software Tools (For Technicians)**
 - This is used when Method 1 fails or when you need to remove the lock *without* wiping data (very advanced).
 - **For MTK Phones (Tecno/Infinix):**
 1. Open **SP Flash Tool** or **Miracle Box**.
 2. Go to the "Format" or "Reset" tab.
 3. Select "Format (Wipe Data)" or "Reset User Lock."
 4. Connect the switched-off phone and start the process. This formats the USERDATA partition of the phone's memory.
 - **Bypassing FRP:**
 - This is a key job for technicians.
 - Customers often forget their Google account after a reset.
 - Tools like **Miracle Box** or **UMT Dongle** have a specific button (e.g., "Reset FRP") that removes this lock.

Evaluation:

1. What is the difference between a SIM lock and a screen lock?
2. What is "FRP," and what is its main purpose?
3. What is the standard, built-in method for removing a forgotten pattern?
4. What is the *consequence* of performing a Factory Reset? (Data loss).
5. What software tool is commonly used to "Reset FRP" on MTK phones?

Assignment:

1. List 4 types of security locks.
2. A customer forgets their pattern. Describe the steps to perform a "Factory Reset" from Recovery Mode.
3. What problem will the customer face *after* the factory reset if they also forgot their Google account?

4. What does "FRP" stand for?
5. Why is "bypassing FRP" a major service offered by GSM technicians?



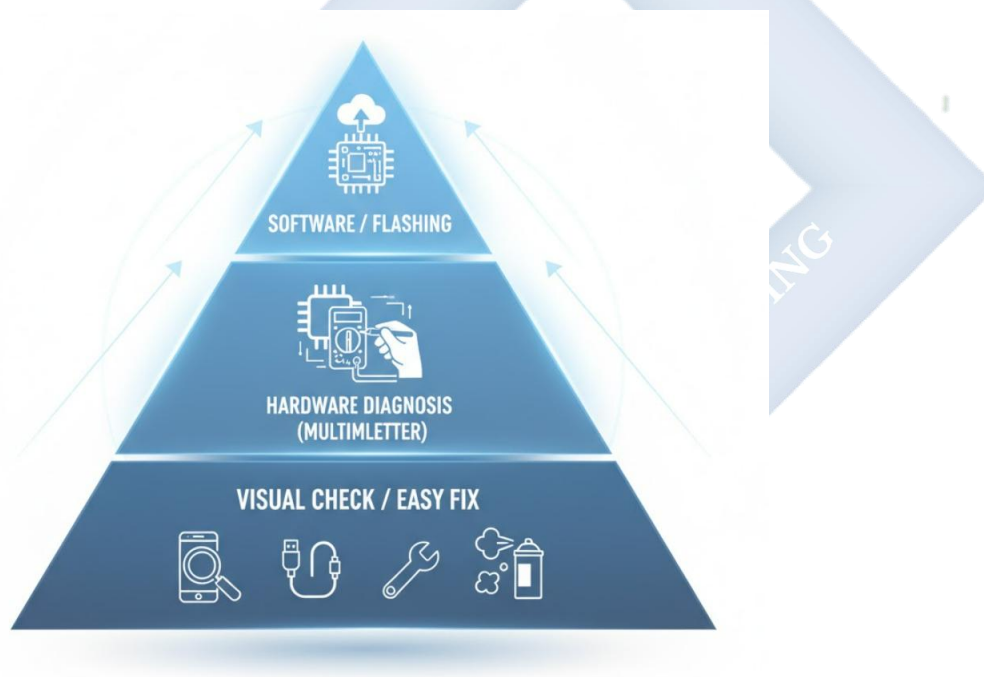
WEEK 3: VIRUS AND MALWARE REMOVAL

Topic: Virus and Malware Removal

Content: Meaning, types, and removal of malware from devices

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "malware," "virus," "adware," and "spyware."
2. List at least four symptoms of a malware-infected phone.
3. Describe the three levels of malware removal (from simple to complex).
4. Explain how to use "Safe Mode" as a diagnostic tool.
5. Identify "flashing" as the final method for removal.



Lesson Content:

1. Meaning and Types of Malware

- **Malware (Malicious Software):** The *general* name for *any* software designed to cause damage or steal information.
- **Types of Malware on Phones:**

1. **Virus:** A program that copies itself and spreads, corrupting data and slowing the phone down. (Less common on phones).
2. **Adware (Annoying):** The *most common* type. Software that automatically displays or downloads *unwanted advertisements* (e.g., pop-up ads on your home screen).
3. **Spyware (Dangerous):** Software that *secretly* records your information (passwords, bank details, messages) and sends it to a hacker.
4. **Ransomware (Very Dangerous):** Software that *encrypts* (locks) all your personal files (photos, documents) and demands a *ransom* (money) to unlock them.

2. Symptoms of a Malware-Infected Phone

- The phone is suddenly very **slow** or **freezes**.
- The battery **drains** extremely fast.
- Your data (airtime or mobile data) is being **used up** quickly.
- You see **pop-up ads** on your home screen or lock screen (Adware).
- New **apps appear** that you did not install.
- The phone **overheats** even when not in use.

3. How Phones Get Infected

- Downloading "cracked" or unofficial apps from outside the Google Play Store / Apple App Store.
- Clicking on suspicious links in emails, WhatsApp, or SMS ("phishing").
- Visiting unsafe websites.

4. Methods of Malware Removal (Step-by-Step)

- **Level 1: The Simple Fix (Antivirus & Uninstall)**
 1. Go to the Google Play Store and install a trusted antivirus app (e.g., Malwarebytes, Avast, Bitdefender).
 2. Run a "Full Scan." The app will find and help you uninstall the malicious app.

3. Manually go to Settings > Apps and look for any strange apps you don't recognize. Uninstall them.
- **Level 2: The "Safe Mode" Fix**
 - **What is Safe Mode?** A diagnostic mode that starts the phone with *all third-party apps disabled*.
 - **How it helps:** If the pop-up ads *stop* in Safe Mode, it proves the problem is caused by an *app* you downloaded.
 - **Procedure:**
 1. Boot the phone into Safe Mode (e.g., hold Power, then long-press the "Power Off" option).
 2. The phone restarts (with "Safe Mode" written on the screen).
 3. Go to Settings > Apps and uninstall the suspicious apps one by one.
 4. Reboot the phone normally.
 - **Level 3: The "Nuclear" Fix (Factory Reset)**
 - If the virus is too deep, the simplest way to remove it is a **Factory Reset (Week 2)**.
 - This erases *everything*, including the virus. The user *must* back up their data (photos, contacts) first.
 - **Level 4: The Technician's Fix (Flashing)**
 - Some malware can even survive a factory reset (by installing itself in the *system* partition).
 - The *only* way to remove this is to **Flash the Phone (Week 1)**. This wipes the *entire* system and installs a 100% clean, new OS.

Evaluation:

1. What is "malware"?
2. What is "adware"? What is the main symptom?
3. What is "spyware"?
4. List three symptoms of an infected phone.
5. What are the three levels of malware removal (Simple, Safe Mode, Factory Reset)?

Assignment:

1. Define Malware, Adware, and Spyware.
2. What is the main way a phone gets infected with a virus?
3. What is "Safe Mode," and how does it help you find a bad app?
4. What is the *final* and most powerful method to remove a very deep virus?
5. A customer has pop-up ads on their home screen. What type of malware is this?



WEEK 4: DATA RECOVERY

Topic: Data Recovery

Content: Causes of data loss; Tools and steps in data recovery

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "data recovery."
2. List at least four common causes of data loss.
3. Explain the *first and most important rule* when data is lost.
4. Identify software tools used for data recovery.
5. Explain why data recovery is not always possible.



Lesson Content:

1. What is Data Recovery?

- **Definition:** Data recovery is the process of *rescuing* (retrieving) inaccessible, lost, corrupted, or deleted data from a storage device (like a phone's internal memory or an SD card).
- **"Deleted" data** is not actually gone. When you "delete" a photo, the phone's OS just *marks the space* as "available to be overwritten." The photo is still there until a *new* file (like a new photo or app) is saved on top of it.

2. Common Causes of Data Loss

1. **Accidental Deletion:** This is the most common. (e.g., "I accidentally deleted all my photos!").
2. **Factory Reset:** Wiping the phone for a password or virus.
3. **Physical Damage:** The phone is broken (e.g., dropped, smashed).
4. **Water Damage:** Water gets on the PCB and causes a short circuit, corrupting the memory chip.
5. **Software Corruption:** A failed update or a virus "bricks" the phone, making data unreadable.

3. The *First and Most Important Rule* of Data Recovery

- **STOP USING THE DEVICE IMMEDIATELY!**
- As explained above, "deleted" data is waiting to be overwritten.
- If you delete a photo and then *continue* to use the phone (take new photos, browse the internet, download apps), the phone will *save new data* on top of the old, deleted photo.
- Once it is overwritten, the data is **gone forever**.
- **Action:** Turn on **Flight Mode** and **do not** use the phone until you can run a recovery tool.

4. Data Recovery Tools and Steps Recovery is usually done with a PC.

- **Tools (PC Software):**

- EaseUS MobiSaver
- Dr. Fone - Data Recovery
- Disk Digger (an app that can run on the phone, but less effective).

- **General Steps:**

1. **Install** the recovery software (e.g., Dr. Fone) on your *computer*.
2. **Connect** the phone to the computer with a USB cable.
3. **Enable "USB Debugging"** on the phone (a special developer mode).
4. **Run the Software:** Launch the tool on your PC.
5. **Scan:** Select the type of data to recover (e.g., "Photos," "Contacts") and start the scan. The tool will scan the phone's "free space" for deleted files.
6. **Preview and Recover:** The tool will show you a preview of what it found. You can select the files and save them to your *computer*.

5. Why Recovery is Not Always Possible

1. **Overwritten Data:** The user kept using the phone, and the data was overwritten.
2. **Encryption:** Most modern phones (iPhones and many Androids) are *encrypted*. This means all the data is scrambled. If the phone is factory reset, the "key" to unscramble the data is destroyed, making recovery *impossible*.
3. **Physical Damage:** If the *memory chip (eMMC)* itself is cracked or destroyed, the data is physically gone.

Evaluation:

1. What is "data recovery"?
2. What is the *first and most important* rule after you accidentally delete a file?
3. List three common causes of data loss.
4. Why is it *bad* to keep using a phone after you delete a photo?
5. Name one software tool used for data recovery.

Assignment:

1. Define data recovery.
2. Explain (in your own words) what "overwritten" means.
3. List the 5 steps for using a PC-based recovery tool.
4. What is "encryption," and why does it make data recovery almost impossible after a factory reset?
5. A customer deletes their photos. What is the *first* thing you should tell them to do?



WEEK 6: PHONE TESTING AND QUALITY CONTROL

Topic: Phone Testing and Quality Control

Content: Testing repaired phones before delivery; Ensuring quality

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Quality Control (QC)."
2. Explain the importance of QC in a repair business (reputation, profit).
3. List at least 10 items on a "Post-Repair QC Checklist."
4. Explain the difference between testing a *hardware* repair and a *software* repair.
5. Demonstrate how to test a screen, speaker, and microphone.



Lesson Content:

1. What is Quality Control (QC)?

- **Definition:** Quality Control is the *process* of testing a product (or in our case, a *service*) to ensure it meets a high standard of quality *before* it is delivered to the customer.
- **Goal:** To find any faults *before* the customer does.
- **The Rule:** *Never* give a phone back to a customer without testing it completely.

2. Importance of Quality Control

1. **Prevents "Comebacks":** A "comeback" is when a customer returns with the *same* or a *new* problem. (e.g., "You fixed my screen, but now my speaker doesn't work!"). This is a disaster for a business.
2. **Saves Money:** A "comeback" means you have to do the job *twice* but only get paid *once*. You lose time and profit.
3. **Builds Reputation (Trust):** When a customer gets their phone back and *everything* works perfectly, they *trust* you. They will come back, and they will tell their friends.
4. **Finds New Faults:** Sometimes, a drop that cracks a screen *also* damages the microphone. QC helps you find this *before* the customer does.

3. The Post-Repair Quality Control (QC) Checklist *After every repair, you must test all major functions, even the ones you didn't repair. (This is because you may have accidentally disconnected a cable or component during the repair)*

A. Hardware Tests:

- [] **Screen:** Is the brightness good? Are there any dead pixels or lines?
- [] **Touch / Digitizer:** Open a drawing app or the keyboard. Test *every part* of the screen. Does it touch accurately?
- [] **Earpiece Speaker:** Make a test call. Can you hear?
- [] **Microphone:** Make a test call (or use the Voice Recorder app). Can you be heard clearly?

- [D] **Loudspeaker:** Play a YouTube video or a ringtone. Is it loud and clear?
- [] **Buttons:** Test Power button, Volume Up, Volume Down. Do they "click"?
- [] **Charging:** Plug in a charger. Does the charging symbol appear?
- [] **SIM Card:** Does the phone detect the SIM? Does it get a network signal?
- [] **Cameras:** Open the camera app. Test *both* the front and rear cameras. Do they focus?
Does the flash work?
- [] **WiFi / Bluetooth:** Connect to a WiFi network.
- [] **Sensors:**
 - **Proximity Sensor:** Make a call and put your hand over the top of the phone. The screen should go black. (This stops your face from touching buttons).

B. Software Tests:

- (After flashing or virus removal)
- [] Is the phone activated? (No FRP lock).
- [] Are all basic apps working (Phone, Camera, Messages)?
- [] Is it connected to the internet?

Evaluation:

1. What is "Quality Control (QC)"?
2. Why is QC important for a technician's reputation?
3. What is a "comeback," and why is it bad for business?
4. List 5 things you must check on a QC checklist after a screen replacement.
5. How do you test the "Proximity Sensor"?

Assignment:

1. Define Quality Control.
2. Explain (in your own words) why QC is important for saving money.
3. Create a "Post-Repair QC Checklist" with at least 10 items.
4. You just replaced a charging port. What 3 *other* things should you test that are *near* the port? (Hint: Microphone, Loudspeaker, Network Antenna).

5. (Practical) Pick up your phone. Go through the QC checklist. Test *every* function on it right now (make a test call, record your voice, test both cameras, test WiFi, test the proximity sensor).



WEEK 8: RECORD KEEPING AND REPORTING

Topic: Record Keeping and Reporting

Content: Writing job completion reports; Daily repair log

Performance Objectives: By the end of this lesson, students should be able to:

1. Review the three main types of records from Term 2 (Repair Log, Income, Expenses).
2. Define "Job Completion Report."
3. Explain the purpose of a Job Completion Report (for the customer and the business).
4. List the 5 key sections of a Job Completion Report.
5. Practically fill out a job log and a completion report for a repair.



Lesson Content:

1. Review of Bookkeeping (from Term 2)

- Bookkeeping is *tracking all transactions*.

- We use three main books:
 1. **Repairs Log (Job Log):** Tracks all *phones* that enter the shop.
 2. **Income Ledger:** Tracks all *money coming IN*.
 3. **Expenses Ledger:** Tracks all *money going OUT*.
- This lesson focuses on *combining* the Job Log with a professional report.

2. What is a Job Completion Report (or "Service Report")?

- **Definition:** This is a simple, professional document (or a stamp/sticker) that you give to the customer *with* their repaired phone.
- It is a *summary* of the repair. It is proof of the work you did.

3. Importance of a Job Completion Report

- **For the Customer (Professionalism & Trust):**
 - It shows them *exactly* what they paid for.
 - It acts as their **receipt** and **warranty card**.
 - It makes you look like a professional, organized business, not an amateur.
- **For the Business (Record Keeping):**
 - It is your *copy* of the job, linked to the Repairs Log.
 - If the customer comes back for a *warranty* repair, you can check your copy of the report to see what was done and if it's still under warranty.

4. Key Sections of a Job Completion Report This can be a pre-printed form or a simple note.

1. **Customer & Phone Details:**
 - Customer Name: (e.g., Mr. John)
 - Phone Model: (e.g., Tecno Pova)
 - Date In: (e.g., Oct 28)
2. **Fault Reported by Customer:**
 - (What the customer *said* was wrong).
 - *Example:* "Customer said the screen was cracked and touch was not working."
3. **Technician's Diagnosis:**

- (What you *found* was wrong).
- *Example:* "Found cracked digitizer. Screen (LCD) is okay. Battery is good."

4. **Work Done (The Repair):**

- (What you *did* to fix it).
- *Example:* "Replaced the screen and digitizer assembly (original part)."
- *Example:* "Cleaned charging port. Tested OK."

5. **Warranty & Final Cost:**

- *Example:* "Total Cost: N8,000"
- *Example:* "Warranty: 30 days on the new screen (does not cover new cracks or water damage)."

5. How it Works with the Repair Log The **Repair Log (Week 4, Term 2)** is your *internal* book. The **Job Completion Report** is the *customer's* copy.

- You can assign a **Job ID** (e.g., #001, #002) to each phone.
- The Repair Log has Job ID #001.
- The Job Completion Report also has Job ID #001.
- This way, you can easily find the job in your records.

Evaluation:

1. What are the three main record books for a technician?
2. What is a "Job Completion Report"?
3. Who is the report for? (The customer).
4. List three key sections of a good completion report.
5. Why is this report important for handling *warranty* claims?

Assignment:

1. Define "Job Completion Report" and "Repairs Log."
2. List the 5 key sections of a Job Completion Report.

3. (Practical) A customer comes in with an iPhone X. The fault is "not charging." Your diagnosis is a "dirty port." The work done is "cleaned port." The price is N500.
 - **Write a Job Completion Report** for this repair.
4. (Practical) A customer comes with a Tecno Spark. The fault is "stuck on logo." The work done is "Flashed phone with new firmware." The price is N2,000.
 - **Write a Job Completion Report** for this repair.
5. Why is it important to write *exactly* what part you replaced (e.g., "original part" vs. "copy part") on the report?



WEEK 9: PRACTICAL: FULL REPAIR CYCLE

Topic: Practical: Full Repair Cycle

Content: Receiving a faulty phone, diagnosing, repairing, and documenting the process

Performance Objectives: By the end of this lesson, students should be able to:

1. Synthesize all skills from all three terms (Hardware, Software, Business).
2. Role-play the "Customer Service" step of receiving a phone.
3. Role-play the "Bookkeeping" step of logging a repair.
4. Role-play the "Diagnostic" step (Hardware or Software).
5. Role-play the "QC and Reporting" step of completing the job.

The 8-Step Flow



Lesson Content:

1. Goal of the "Full Cycle" The goal of this lesson is to *combine* everything we have learned into a single, professional process. This is what a real technician does every day.

2. The 8 Steps of the Full Repair Cycle (This combines Term 1, Term 2, and Term 3)

- **Step 1: Receive the Phone (Customer Service - Term 2)**
 - **Action:** Greet the customer ("Welcome!"), listen actively to their problem, and show empathy.
 - **Ethics:** Protect their privacy from the start.
- **Step 2: Log the Job (Bookkeeping - Term 2 / Term 3)**
 - **Action:** Take the customer's name, number, and phone model. Write it in the **Repairs Log**.
 - **Action:** Write down the *fault* they described.
 - **Action:** Give the customer a "job tag" or "receipt" with a Job ID number.
- **Step 3: Diagnose the Fault (Hardware - Term 1)**
 - **Action:** Start with the *simple* things first.
 - *Visual:* Check for water damage, cracks, or dirt in ports.
 - *Test:* Use a known-good charger.
 - *Internal:* Open the phone (ESD-safe!). Use the **Multimeter**.
 - Test the **Battery Voltage**.
 - Test the **Power Button (Continuity)**.
 - Check for short circuits.
- **Step 4: Give the Quote (Customer Service - Term 2)**
 - **Action:** *Before* you repair, *call* the customer.
 - **Action:** Explain the fault *simply* ("You need a new battery"), state the *price* ("It will be N4,000"), and state the *time* ("It will be ready this afternoon").
 - **Action:** Get their *permission* to proceed.
- **Step 5: Perform the Repair (Hardware/Software - Term 1 & 3)**
 - **Action:** Get the new part.

- **Action:** (Hardware) Safely disassemble, replace the part (e.g., screen, port), and reassemble.
- **Action:** (Software) If it's a software fault, connect to the PC and **Flash** or **Unlock** the phone.
- **Step 6: Quality Control (QC) (Testing - Term 3)**
 - **Action:** This is the *most critical* step.
 - **Action:** Go through your **QC Checklist**. Test *everything* (screen, touch, cameras, speakers, mic, network, charging, buttons).
- **Step 7: Document the Job (Reporting - Term 3)**
 - **Action:** Write out the **Job Completion Report** (for the customer).
 - **Action:** Update your **Income Ledger** (with the price) and **Expenses Ledger** (with the cost of the part you used).
- **Step 8: Deliver to Customer (Customer Service - Term 2)**
 - **Action:** Call the customer to tell them the phone is ready.
 - **Action:** When they arrive, *show* them the repair is working.
 - **Action:** Give them the phone and the Job Completion Report (receipt).
 - **Action:** Thank them for their business.

Evaluation: (Observation-based)

1. Divide the class into groups and give them a *faulty phone* (e.g., one with a loose battery connector).
2. Have them *perform* the 8-step cycle.
3. *Assess:* Did they log the job? Did they diagnose with a multimeter? Did they do a QC test *after* the repair? Did they fill out the report?

Assignment:

1. List the 8 steps of the "Full Repair Cycle" in the correct order.
2. Why is "Step 4: Give the Quote" done *before* "Step 5: Perform the Repair"?
3. Why is "Step 6: Quality Control" done *before* "Step 8: Deliver to Customer"?

4. (Practical) You get a phone with a broken screen. Write a 5-line script of what you would say to the customer (Greeting, Diagnosis, Quote).
5. What two (2) records must you update *after* the job is finished and the customer has paid?



Week 10: REVISION

Topic: Revision

Content: Comprehensive review of all topics covered

Performance Objectives: By the end of this lesson, students should be able to:

1. Verbally define all key software tools and their uses.
2. Verbally explain the difference between a Factory Reset, Flashing, and FRP Bypass.
3. Verbally describe the 8-step "Full Repair Cycle."
4. Review all key business concepts (Bookkeeping, Customer Service, Ethics).
5. Review all key hardware concepts (ESD, Multimeter, Components).

Lesson Content:

1. Summary of Term 3 Topics (The "Software & Process" Term) This term focused on advanced *software* repairs and the professional *process* of running a business.

- **Week 1 (Software Tools):**
 - Tools like **SP Flash Tool** (for MTK) and **Odin** (for Samsung) are used to flash firmware.
 - Tools like **Miracle Box** are professional "all-in-one" dongles.
 - **Drivers** must be installed first.
- **Week 2 (Unlocking):**
 - **Factory Reset:** Wipes data, removes screen lock. Done from **Recovery Mode**.
 - **FRP (Factory Reset Protection):** The Google lock *after* a reset.
 - Software tools are used to **Bypass FRP** or **Reset User Locks**.
- **Week 3 (Malware):**
 - **Malware:** Adware (pop-ups), Spyware (steals info), Ransomware (locks files).
 - **Removal:** 1. Antivirus App, 2. Safe Mode, 3. Factory Reset, 4. Flashing (final solution).
- **Week 4 (Data Recovery):**

- **Rule #1: STOP USING THE PHONE.** (To prevent overwriting).
- Use PC tools (like Dr. Fone) to scan for deleted files.
- Often *impossible* due to **Encryption**.
- **Week 6 (Quality Control - QC):**
 - Testing *all* functions (screen, touch, camera, mic, etc.) *after* a repair.
 - Prevents "comebacks" and builds reputation.
- **Week 8 (Reporting):**
 - A **Job Completion Report** is the *customer's receipt* and *warranty card*. It details the fault, the fix, and the price.
- **Week 9 (Full Cycle):**
 - The 8-step process: Receive -> Log -> Diagnose -> Quote -> Repair -> Test (QC) -> Document -> Deliver.

2. Class Activity: Final Q&A (Review of All Terms) (Teacher asks rapid-fire questions from all three terms)

- "What is ESD?" (Term 1)
- "What tool tests a battery?" (Term 1)
- "What does a 'beep' mean in a continuity test?" (Term 1)
- "What is a 'growth mindset'?" (Term 2)
- "What is 'reinvestment'?" (Term 2)
- "What does L.E.A.F. stand for?" (Term 2)
- "What is 'FRP'?" (Term 3)
- "What tool flashes a Tecno phone?" (Term 3)
- "What is the first thing you do if you delete a photo?" (Term 3)

Evaluation: This week's evaluation is the Third Term (Final) Examination.

Assignment:

1. Study all notes from all three terms.
2. Make a list of 5 Hardware tools and 3 Software tools.
3. Make a list of 5 Business terms (e.g., Profit, Asset).

4. Write down the 8 steps of the "Full Repair Cycle."
5. What do you think is the *most important* skill for a GSM technician: (a) Hardware skill, (b) Software skill, or (c) Customer Service skill? Explain your answer in two sentences.

